

# RL-UDHFL: Reinforcement Learning-Enhanced Utility-Driven Hierarchical Federated Learning for IoT

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**Abstract**—Decentralized federated learning (DFL) is recognized as a key paradigm for training models in resource-constrained, privacy-sensitive Internet of Things (IoT) environments. However, its real-world deployment is hindered by device heterogeneity, limited resources, and unpredictable node trustworthiness. To address these challenges, an innovative framework, namely reinforcement learning-driven utility-based decentralized hierarchical federated learning (RL-UDHFL), is proposed, in which reinforcement learning (RL) is leveraged for adaptive optimization across three tiers: edge, coordination, and global aggregation. At the edge, participants are selected through an RL-based adaptive utility-driven participant selection (RL-AUDPS) mechanism, based on a utility function that accounts for computational resources, energy, data quality, and reputation. At the coordination level, self-tuning adaptive clustering is applied, and a trust-aware gossip protocol (TAGS) is employed to enable robust intercluster communication. At the global level, the reputation-based weighting is utilized, and on-the-fly anomaly detection is performed to ensure model integrity. Through extensive simulations, it is demonstrated that an RL-UDHFL achieves a model accuracy of 98%, surpassing hierarchical benchmarks, such as HAFedRL (93.5%) and T-FedHA (92%). This superior performance is attributed to the framework's capability to balance high accuracy, efficient resource utilization, and system reliability, thereby providing a scalable and robust blueprint for deploying sustainable and trustworthy learning systems in complex IoT applications.

**Index Terms**—Adaptive utility-driven participant selection (AUDPS), Internet of Things (IoT), reinforcement learning (RL), RL-driven utility-based decentralized hierarchical federated learning (RL-UDHFL).

## NOMENCLATURE

|     |                                                   |
|-----|---------------------------------------------------|
| $N$ | Total number of IoT devices in the system.        |
| $K$ | Number of clusters formed by adaptive clustering. |

Received 8 August 2025; revised 9 October 2025, 13 November 2025, and 4 December 2025; accepted 11 December 2025. Date of publication 19 December 2025; date of current version 23 February 2026. (*Corresponding author: Seyedakbar Mostafavi.*)

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Digital Object Identifier 10.1109/JIOT.2025.3646594

|                                           |                                                                    |
|-------------------------------------------|--------------------------------------------------------------------|
| $\mathcal{C}_k$                           | $k$ th cluster, containing a subset of IoT devices.                |
| $\mathcal{D}_i$                           | Local dataset of device $i$ .                                      |
| $ \mathcal{D}_i $                         | Number of samples in $\mathcal{D}_i$ .                             |
| $f_w(\cdot)$                              | Model parameterized by weights $w$ .                               |
| $\ell(f_w(x), y)$                         | Task-specific loss for input $x$ and label $y$ .                   |
| $\mathcal{L}_i(w)$                        | Local empirical loss of device $i$ .                               |
| $\mathcal{L}(w)$                          | Global aggregated loss function.                                   |
| $U(w)$                                    | Utility function (accuracy-driven).                                |
| $\pi$                                     | RL-based device selection policy.                                  |
| $\lambda_1, \lambda_2$                    | Trade-off coefficients for communication and energy costs.         |
| $\text{CommCost}(\pi)$                    | Communication cost under policy $\pi$ .                            |
| $\text{EnergyCost}(\pi)$                  | Energy consumption under policy $\pi$ .                            |
| $R_i$                                     | Reputation score of device $i$ ( $0 \leq R_i \leq 1$ ).            |
| $CS_i^t$                                  | Contribution score of device $i$ at round $t$ .                    |
| $Q_i^t$                                   | Update quality based on cosine similarity of gradients.            |
| $\text{TI}_i^t$                           | Timeliness indicator (1 if on-time, else 0).                       |
| $w_q, w_t$                                | Weights for quality and timeliness components ( $w_q + w_t = 1$ ). |
| $\gamma_r, \delta_r$                      | Growth and decay factors for reputation update.                    |
| $\theta_c$                                | Threshold for determining positive contribution.                   |
| $D_i$                                     | Feature vector of device $i$ .                                     |
| $C_j$                                     | Centroid vector of cluster $j$ .                                   |
| $W_{ik}$                                  | Reliability weight of $k$ th feature of device $i$ .               |
| $d_w(i, j)$                               | Weighted distance between device $i$ and cluster $j$ .             |
| $\mathcal{C}_{\text{comp}}(C)$            | Cluster compactness metric (lower is better).                      |
| $\mathcal{C}_{\text{sep}}(C)$             | Cluster separation metric (higher is better).                      |
| $\alpha_{\text{adp}}, \beta_{\text{adp}}$ | RL-learned adaptive clustering parameters.                         |
| $K_{\text{new}}$                          | Updated number of clusters (14).                                   |
| $\tau_{\text{ie}}, \Delta_{\text{ie}}$    | Actual and maximum allowed communication delay.                    |
| $B_r$                                     | Bound on local update magnitude for convergence stability.         |

|                                                                    |                                                               |
|--------------------------------------------------------------------|---------------------------------------------------------------|
| $c_j^t$                                                            | Device assignment set for cluster $j$ at iteration $t$ .      |
| $L(C^t)$                                                           | Objective function value at iteration $t$ .                   |
| $U_i$                                                              | Composite utility score of device $i$ .                       |
| $C_i, B_i, Q_i, R_i$                                               | Computational, battery, data quality, and reputation factors. |
| $w_C, w_B, w_Q, w_R$                                               | Adaptive utility weights ( $w_C + w_B + w_Q + w_R = 1$ ).     |
| $P_i$                                                              | Selection probability of device $i$ .                         |
| $n_s$                                                              | Number of selected participants per cluster.                  |
| $\rho_s$                                                           | Base selection ratio for each cluster.                        |
| $\delta_a$                                                         | Adaptive scaling coefficient for quota adjustment.            |
| $R_{\text{available}}, R_{\text{max}}$                             | Available and maximum resource capacity.                      |
| $S_t, A_t, R_t$                                                    | State, action, and reward at time $t$ in the RL agent.        |
| $\pi_{\text{RL}}$                                                  | Learned RL policy for participant selection.                  |
| $\eta_{\text{global}}$                                             | Global learning rate in model aggregation.                    |
| $\lambda_{\text{acc}}, \lambda_{\text{en}}, \lambda_{\text{comm}}$ | Reward balancing coefficients.                                |
| $\gamma_i$                                                         | Device-specific decay factor in TA-RMSProp.                   |
| $\gamma_{\text{base}}, \gamma_{\text{max}}$                        | Minimum and maximum RMSProp decay bounds.                     |
| $E[g_t^2]_i$                                                       | Moving average of squared gradients for device $i$ .          |
| $\eta / \sqrt{E[g_t^2]_i + \epsilon}$                              | Adaptive per-device learning rate.                            |
| $g_{\text{min}}, g_{\text{max}}$                                   | Lower/upper bounds of gradient magnitude.                     |
| $\epsilon_{\text{conv}}$                                           | Convergence tolerance threshold.                              |
| $T_{\text{adapt}}$                                                 | Interval for triggering adaptive cluster update.              |
| $\pi^*$                                                            | Optimal learned RL policy maximizing long-term utility.       |

## I. INTRODUCTION

THE widespread adoption of the Internet of Things (IoT) devices in diverse domains, such as smart cities, industrial automation, autonomous vehicles, and healthcare is generating an unprecedented volume of data at the network edge [1], [2], [3], [4]. This rapid data proliferation necessitates intelligent, on-device machine learning (ML) capabilities [1], [2]. The traditional centralized ML paradigms are often impractical for these environments due to significant challenges, including stringent data privacy regulations, limited communication bandwidth, and the severe resource constraints of the IoT devices [5], [6]. Federated learning (FL) has emerged as a prominent distributed learning paradigm to address these issues by enabling collaborative model training locally on devices without sharing raw sensitive data [7], [8], [9], [10].

Despite the promising advantages of FL, many existing decentralized FL (DFL) and hierarchical FL (HFL) frameworks still face significant challenges in real-world deployments [11], [12]. They often lack mechanisms to autonomously adapt to the highly dynamic nature of IoT

networks, which includes fluctuating device availability, intermittent connectivity, and heterogeneous resource capacities. Furthermore, managing non-independent and identically distributed (non-IID) data and ensuring robustness against unreliable or malicious participants remain critical unresolved challenges. These issues can severely degrade model performance and stability, making them significant hurdles in scaling FL systems for large-scale, heterogeneous IoT applications.

This article introduces an efficient and robust hierarchical framework for FL specifically designed to address these operational challenges in dynamic IoT systems. Unlike traditional approaches that propose new core FL algorithms, our primary contribution is a system architecture that integrates intelligent adaptive mechanisms to optimize the entire FL process. The proposed architecture is structured into three tiers—edge, coordination, and global aggregator—creating a scalable, efficient, and resilient operational environment for the IoT networks.

The core of our framework is an RL-AUDPS. An RL agent dynamically learns to select the most suitable participants for each training round by optimizing a multiobjective utility function. This function evaluates real-time device metrics, including computational resources, energy status, data quality, and dynamic reputation score. This intelligent participant selection is complemented by a self-tuning adaptive clustering mechanism, which effectively manages the data heterogeneity across devices, and a trust-aware gossip protocol (TAGS) that ensures efficient, reliable communication between clusters. Furthermore, the global level is fortified with reputation-based weighting and on-the-fly anomaly detection to secure the learning process against unreliable data and malicious participants.

### A. RQs and Contributions

This article addresses the following pivotal research questions (RQs), motivated by gaps identified in the literature on HFL for IoT environments, as follows.

- 1) *RQ1*: How can reinforcement learning (RL) be leveraged to dynamically optimize participant selection in HFL frameworks operating in heterogeneous, resource-constrained IoT environments?
- 2) *RQ2*: What is the impact of integrating adaptive clustering and trust-aware gossip communication on the robustness, scalability, and reliability of HFL in practical IoT deployments?
- 3) *RQ3*: To what extent can the proposed framework mitigate the effects of device heterogeneity and non-IID data distributions to improve model accuracy, convergence stability, and communication efficiency?

#### Contributions:

To directly address these RQs, this work makes the following contributions.

- 1) *Hierarchical Architecture (RQ2 and RQ3)*: We propose and validate a three-tier hierarchical architecture that enhances scalability, optimizes resource allocation, and reduces communication overhead in IoT environments. This design addresses practical challenges, such as intermittent connectivity and device heterogeneity.

- 2) *RL-AUDPS Mechanism (RQ1 and RQ3)*: We introduce an RL-based adaptive utility-driven participant selection (RL-AUDPS) mechanism. It dynamically selects participants based on computational resources, energy availability, data quality, and trustworthiness, improving overall system robustness and convergence efficiency.
- 3) *Security via Trust and Anomaly Detection (RQ2 and RQ3)*: We integrate a multilevel reputation system and anomaly detection framework to safeguard the FL process, ensuring model integrity even in the presence of unreliable or malicious participants.
- 4) *Comprehensive Evaluation (RQ1–RQ3)*: We rigorously evaluate the proposed framework across multiple datasets, realistic IoT network scenarios, and diverse performance metrics. The results demonstrate significant improvements in model accuracy, communication efficiency, and energy consumption compared to state-of-the-art HFL methods.

The remainder of this article is organized as follows. Section II reviews the related work. Section IV presents the proposed framework. Section V-C details the experimental setup and results. Finally, Section VI concludes this article and outlines the future directions.

## II. RELATED WORKS

FL is a distributed ML paradigm that facilitates collaborative model training without the need for centralized data aggregation. This approach ensures data privacy by allowing devices to train models locally and share only updated parameters. FL is particularly beneficial in privacy-sensitive applications, such as the IoT and healthcare, where data privacy, communication overhead, and limited computational power are significant concerns [11], [12].

### A. CFL and DFL

FL is commonly classified into centralized FL (CFL) and DFL. In CFL, a central server coordinates model aggregation in a star topology [13], [14], [15], ensuring synchronized updates and low latency in small-scale systems. CFL is straightforward to implement, but suffers from limited scalability [16], vulnerability to a single point of failure [17], [18], and a concentrated trust model in which the central server is both security anchor and prime attack target. CFL efficiently handles small updates and mitigates malicious behavior, but its scalability is limited by communication overhead [19]. Conversely, DFL removes the central server, performing collaborative aggregation among peers in either fully distributed or clustered topologies [20]. It supports asynchronous or semisynchronous updates [21], scales well in large dynamic networks [22], and provides distributed fault tolerance. Trust is decentralized [23], enhancing resilience but broadening the attack surface and necessitating advanced trust mechanisms [21], [23]. DFL avoids central bottlenecks [22], yet may incur higher communication overhead due to peer-to-peer exchange, greater synchronization complexity, and variable latency in heterogeneous conditions [21]. CFL is therefore suitable for smaller IoT or healthcare systems, while DFL fits large-scale

IoT, vehicular, and mission-critical industrial networks [24], albeit with increased implementation complexity and privacy-driven trust management challenges.

### B. Comparative Analysis of FL Architectures

FL architectures have evolved from classical centralized methods, such as federated averaging (FedAvg) [25] and federated proximal (FedProx) [26], to adaptive centralized schemes, including heterogeneous FL (HeteFL) [27], adaptive FL (AdaFL) [28], and cluster-driven adaptive training (CATA-Fed) [29]. More advanced frameworks integrate asynchronous FL with optimization mechanisms, e.g., STAR-RIS-aided VRCSL for latency reduction [30] and STAR-RIS-assisted VRCS leveraging quantum federated RL for joint mode selection and resource allocation [31]. In addition, CAMP-GNN [32] introduces a constraint-aware graph neural network for dynamic resource allocation in software projects, combining skill alignment, task urgency, and budget considerations to improve allocation accuracy and reduce cost and deadline violations.

Hierarchical methods, such as those proposed by Qiu et al. [33], Zhang et al. [34], and FedAC [35], employ clustering or scheduling heuristics to enhance privacy, communication efficiency, and latency while reducing the load on aggregators. Specifically, Qiu et al. [33] cluster clients based on data similarity or physical proximity, Zhang et al. [34] optimizes scheduling strategies to balance communication loads, and FedAC [35] applies adaptive intracluster learning for more efficient local model updates and coordination.

RL-based architectures, including hierarchical reinforcement FL (HFRL) [36], [37], and hierarchical adaptive federated RL (HAFedRL) [38], adopt actor-critic or primal-dual approaches to dynamically allocate resources and schedule tasks across multiple hierarchical levels. Trusted hierarchical asynchronous FL (T-FedHA) [39] integrates blockchain-enabled trust mechanisms into hierarchical asynchronous FL, improving fault tolerance and system security, albeit at the cost of additional computational and communication overhead.

Comparative analyses reveal distinct trade-offs. Centralized baselines (FedAvg [25] and FedProx [26]) are simple to implement but suffer from limited scalability and high aggregator load. Adaptive centralized methods (HeteFL [27], AdaFL [28], and CATA-Fed [29]) enhance efficiency and accuracy but remain single-objective and centrally bottlenecked. Heuristic hierarchical methods (Qiu et al. [33], Zhang et al. [34], and FedAC [35]) achieve high communication efficiency and scalability but often overlook energy consumption and trust management. RL-based architectures (HFRL [37] and HAFedRL [38]) provide high adaptability and accuracy but assume cooperative client behaviors and may introduce significant learning complexity. Blockchain-enhanced frameworks (T-FedHA [39]) ensure trust and security yet impose substantial computational and communication overhead. Furthermore, bio-inspired optimization techniques have been investigated for mobility-aware clustering in dynamic multi-UAV IoT networks, demonstrating improved cluster formation and routing efficiency [40].

Our proposed three-tier, utility-driven hierarchical RL framework combines the adaptability of RL-based methods with explicit, adaptive energy awareness and a lightweight reputation-based trust system. It selects participants based on computing capacity, data heterogeneity, trust score, and energy status, achieving high accuracy, strong communication efficiency, excellent scalability, and low aggregator load while maintaining moderate distributed complexity. By avoiding blockchain, it ensures robustness with minimal overhead, making it particularly suitable for large-scale, resource-constrained IoT environments.

### III. SYSTEM MODEL FORMULATION

In this section, we describe the system model for the HFL framework and formulate the problem of client selection based on a utility-driven optimization objective. The goal is to optimize communication cost, energy consumption, and model accuracy by dynamically selecting clients using an RL policy.

#### A. HFL System Model

We consider an HFL system composed of  $N$  heterogeneous IoT devices, partitioned into  $K$  clusters  $\{C_1, \dots, C_K\}$ , with each cluster  $C_k$  consisting of a set of devices coordinated by a global aggregator. Each device  $i$  holds a private dataset  $\mathcal{D}_i$  of size  $|\mathcal{D}_i|$  and performs local training under resource constraints (computational, memory, and energy). We assume that all devices are performing distributed gradient descent with local updates of the model parameters.

Let  $\mathbf{w} \in \mathbb{R}^d$  be the global model weights, where  $d$  is the dimensionality of the model parameters. The local loss function for device  $i$  is calculated as

$$\mathcal{L}_i(\mathbf{w}) = \frac{1}{|\mathcal{D}_i|} \sum_{(x,y) \in \mathcal{D}_i} \ell(f_w(x), y) \quad (1)$$

where  $f_w(x)$  is the model's prediction for input  $x$  with parameters  $w$ , and  $\ell(f_w(x), y)$  is the loss function (e.g., squared loss or cross-entropy) that measures the discrepancy between predicted value  $f_w(x)$  and the true label  $y$ .

The global objective is to minimize the aggregated local losses over all devices, weighted by the size of each device's dataset, which is given by

$$\mathcal{L}(\mathbf{w}) = \sum_{i=1}^N \frac{|\mathcal{D}_i|}{\sum_{j=1}^N |\mathcal{D}_j|} \mathcal{L}_i(\mathbf{w}). \quad (2)$$

To further improve the model's efficiency and scale in heterogeneous environments, we incorporate a dynamic client selection process that balances the energy, communication, and accuracy trade-offs.

#### B. Problem Formulation for Client Selection

We aim to select a subset of devices  $\mathcal{C}_t \subseteq \mathcal{N}$  at each training round to maximize the system's overall utility while minimizing energy consumption and communication cost. We formalize the utility function for the selected set of clients at round  $t$ , denoted as  $\mathcal{C}_t$ , as follows:

$$U(\mathcal{C}_t) = \sum_{i \in \mathcal{C}_t} (\alpha A_i - \beta E_i - \gamma L_i) \quad (3)$$

where  $A_i$  is the expected accuracy of device  $i$ ,  $E_i$  is the energy consumption of device  $i$ ,  $L_i$  is the communication cost of device  $i$ , and  $\alpha$ ,  $\beta$ , and  $\gamma$  are the coefficients that weight the relative importance of each factor in the optimization.

The optimization goal is to select the set  $\mathcal{C}_t$  of clients that maximizes this utility subject to constraints on energy consumption and communication cost. These constraints are formulated as

$$E(\mathcal{C}_t) = \sum_{i \in \mathcal{C}_t} E_i \leq E_{\max}, \quad L(\mathcal{C}_t) = \sum_{i \in \mathcal{C}_t} L_i \leq L_{\max} \quad (4)$$

where  $E(\mathcal{C}_t)$  and  $L(\mathcal{C}_t)$  denote the total energy consumption and communication cost for the selected clients, respectively, while  $E_{\max}$  and  $L_{\max}$  represent the maximum allowable energy and communication costs, respectively.

#### C. Solution Strategy: RL Policy for Client Selection

To solve the client selection problem in real-time, we use an RL approach, where an RL agent learns a policy  $\pi_t$  to select the optimal clients at each training round based on the observed state.

1) *State Representation*: The state  $\mathbf{s}_t$  at time  $t$  is represented as a vector containing key metrics from each device, such as available computational resources, communication cost, energy consumption, and data quality. Formally, the state is represented as

$$\mathbf{s}_t = [R_{\text{cpu}}^t, R_{\text{mem}}^t, R_{\text{energy}}^t, Q_i^t] \quad \forall i \in \mathcal{N} \quad (5)$$

where  $R_{\text{cpu}}^t$  is the available CPU resource at time  $t$ ,  $R_{\text{mem}}^t$  is the available memory at time  $t$ ,  $R_{\text{energy}}^t$  is the remaining energy at time  $t$ , and  $Q_i^t$  is the data quality for device  $i$  at time  $t$ .

2) *Action and Reward Function*: At each time  $t$ , the RL agent selects a set of clients  $\mathcal{C}_t$  based on the observed state. The reward function encourages the selection of clients that maximize the utility function while minimizing communication and energy costs. The reward function is defined as

$$R_t = \alpha \cdot \text{Acc}_t - \beta \cdot \text{CommCost}_t - \gamma \cdot \text{EnergyCost}_t \quad (6)$$

where  $\text{Acc}_t$  is the total accuracy of the selected clients,  $\text{CommCost}_t$  is the total communication cost, and  $\text{EnergyCost}_t$  is the total energy consumption.

The RL agent maximizes this reward to select clients that contribute to the most efficient training process.

3) *Q-Learning for Policy Optimization*: To optimize the client selection process, the RL agent uses Q-learning. The goal of the RL agent is to learn a policy  $\pi^*$  that maximizes the expected cumulative reward over time. The Q value update rule is given by

$$Q(\mathbf{s}_t, \mathcal{C}_t) \leftarrow Q(\mathbf{s}_t, \mathcal{C}_t) + \alpha \left( R_t + \gamma \max_{\mathcal{C}_{t+1}} Q(\mathbf{s}_{t+1}, \mathcal{C}_{t+1}) \right) \quad (7)$$

$$- Q(\mathbf{s}_t, \mathcal{C}_t) \quad (8)$$

where  $\alpha$  is the learning rate,  $\gamma$  is the discount factor, and  $R_t$  is the reward at time  $t$ .

The Q-learning algorithm allows the RL agent to iteratively improve its policy by learning from the rewards it receives based on its actions.



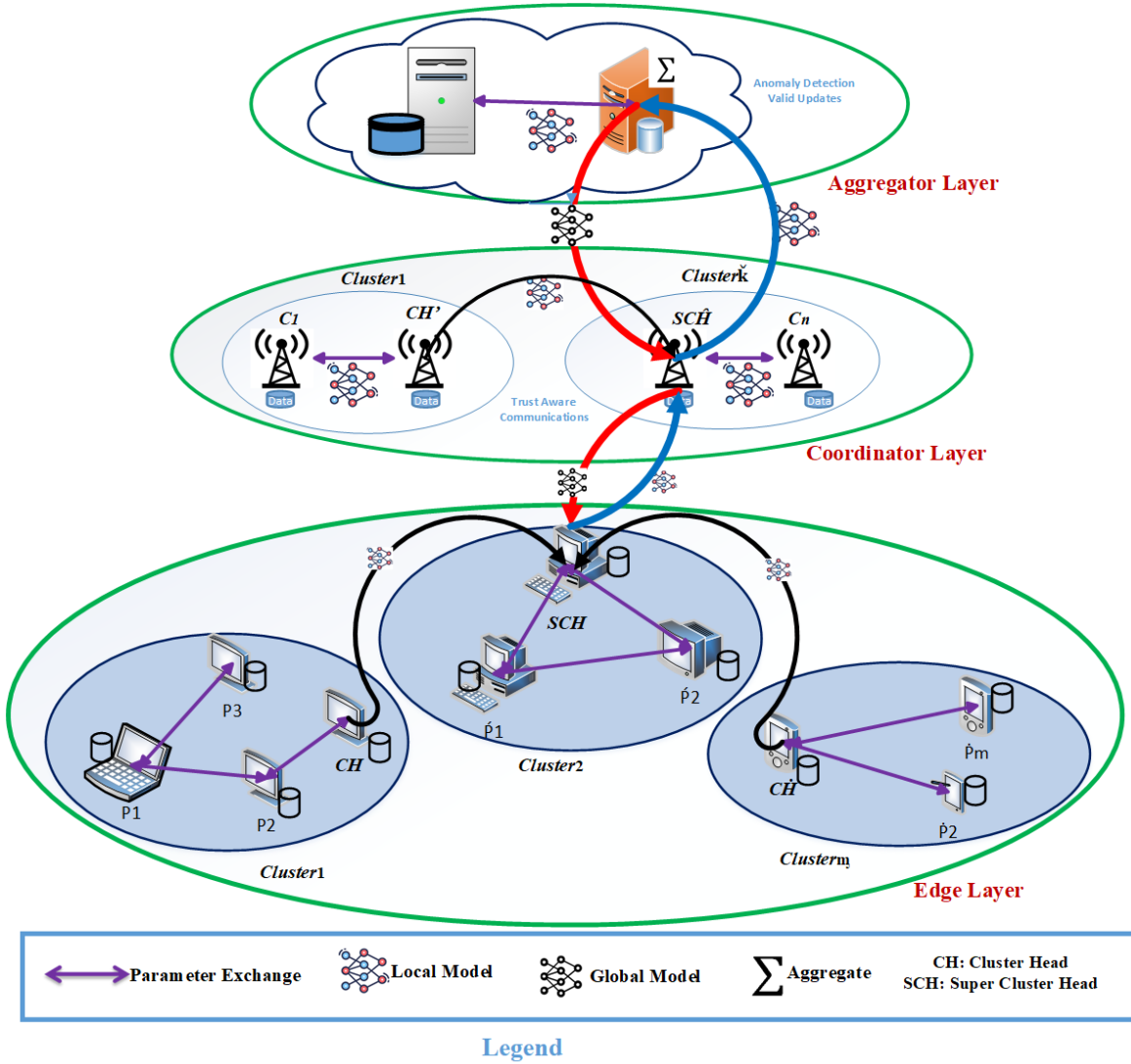


Fig. 1. Overview of the proposed three-tier RL-UDHFL architecture.

#### IV. PROPOSED METHOD

The proposed method employs a hierarchical architecture for FL in IoT environments, consisting of three distinct tiers: the edge level, the coordination level, and the global aggregator. Each tier is assigned specific tasks to optimize communication, coordination, and model convergence while addressing key challenges inherent to IoT systems. To mitigate resource limitations, the proposed method uses adaptive mechanisms at each level to optimize the utilization of computational power, energy, and bandwidth. Device heterogeneity is managed through dynamic client selection and adaptive clustering, allowing devices with varying capabilities to participate effectively without degrading performance. To address non-IID data distributions, the proposed framework incorporates reinforced participant selection strategies that consider data quality, improving model accuracy despite statistical heterogeneity. Communication overhead is significantly reduced by the multitiered structure, which performs local aggregations at the edge and coordination levels before a final global update, thus enhancing scalability. Finally, to ensure secu-

rity against malicious participants, the architecture integrates trust-aware mechanisms, including a reputation-based scoring system for client selection and a dynamic, TAGS for robust communication. Fig. 1 illustrates the overall architecture of the proposed RL-driven utility-based decentralized HFL (RL-UDHFL) framework, comprising the edge layer for local IoT training, the coordination layer for trust-aware intercluster aggregation, and the global aggregator for anomaly detection and weighted global model fusion.

For clarity, the Nomenclature summarizes the main notations used throughout this article, providing a quick reference for the symbols and their meanings.

##### A. Edge Level

The edge level, as the foundation of proposed architecture, comprises numerous IoT devices tasked with local data acquisition and initial processing. These devices typically operate under considerable resource constraints, including limited computational power, memory, and energy, while also exhibiting significant heterogeneity. To mitigate these challenges, the

edge level is equipped with a range of intelligent adaptive mechanisms. These mechanisms are specifically designed to effectively manage the inherent limitations of the devices, optimize local performance, and ensure the reliability and trustworthiness of the contributions before they are transmitted to higher levels for aggregation.

1) *Dynamically Updated Reputation Score Mechanism:*

In this framework, the trustworthiness of each device  $i$  is quantified by a dynamic reputation score, denoted by  $R_i$ . This score, a value between 0 (completely untrustworthy) and 1 (highly trustworthy), represents the device's historical reliability. The reputation score is updated after each communication round—a full cycle of the FL process—based on the quality of the device's contribution during that round.

The model update submitted by device  $i$  to the cluster's centroid for aggregation is evaluated using a contribution score defined as  $CS_i^t = w_q \times Q_i^t + w_t \times T_i^t$ , where  $Q_i^t$  denotes the quality of the update—measured by the cosine similarity between the device's local update vector ( $\Delta w_i$ ) and the aggregated cluster update ( $\Delta w_c$ )—with higher similarity indicating better alignment with the cluster's consensus. The term  $T_i^t$  is a binary timeliness score, set to 1 if the update is submitted within the predefined time window, and 0 otherwise.  $w_q$  and  $w_t$  denote the weighting hyperparameters that sum to 1.

Based on the computed contribution score  $CS_i^t$ , the reputation score  $R_i$  of each client  $i$  is iteratively updated using a rule that rewards positive contributions and penalizes negative ones. A contribution is considered positive if  $CS_i^t$  exceeds a predefined threshold  $\theta_c$ .

- 1) For a positive contribution ( $CS_i^t \geq \theta_c$ ), the reputation score is increased toward 1 using a growth factor  $\gamma_r$

$$R_i^{t+1} = R_i^t + (\gamma_r \times (1 - R_i^t)). \quad (9)$$

- 2) For a negative contribution ( $CS_i^t < \theta_c$ ), the reputation score is penalized toward 0 using a decay factor  $\delta_r$

$$R_i^{t+1} = R_i^t - (\delta_r \times R_i^t). \quad (10)$$

2) *Smart Adaptive Clustering With Self-Tuning K-Value:* To effectively manage device and data heterogeneity, the framework employs a dynamic adaptive clustering mechanism. This mechanism groups IoT devices based on data similarity, real-time resource availability, and a device reputation score. The clustering process is guided by several key metrics designed to evaluate the quality of the cluster formations.

The foundational metric is the weighted distance function, which quantifies the proximity between a device  $i$  and a cluster centroid  $j$ . It incorporates feature-level weights ( $W_{ik}$ ) to emphasize more reliable data sources. This weighted distance, denoted as  $d_W(i, j)$ , is calculated as

$$d_W(i, j) = \sqrt{\sum_k W_{ik} (D_{ik} - C_{jk})^2} \quad (11)$$

where  $D_{ik}$  is the  $k$ th feature of device  $i$ 's data,  $C_{jk}$  is the  $k$ th feature of cluster  $j$ 's centroid, and  $W_{ik}$  is the dynamically determined weight assigned to the  $k$ th feature of device  $i$ , reflecting its importance and, uniquely, its reliability in contributing valid feature data. In this case, we have two scenarios as follows.

- 1) *Cluster Compactness:* ( $C_{\text{comp}}$ ) quantifies the internal cohesion of a cluster, where lower values are better. It reflects how closely devices are grouped around their centroid. For a given set of clusters  $C$ , it is expressed as

$$C_{\text{comp}}(C) = \sum_{c \in C} \frac{1}{|c|} \sum_{i \in c} W_i \left( \frac{1}{|c|} \sum_{p \in c} d_W(p, c)^2 \right) \quad (12)$$

where  $c$  is a cluster in  $C$ ,  $i$  represents a device within that cluster, and  $|c|$  is the total number of devices in cluster  $c$ . The term  $W_i$  is the importance weight assigned to device  $i$  and is defined by the device's reputation score,  $R_i$ , such that  $W_i = R_i$ . A lower compactness value signifies a more cohesive and trustworthy cluster structure.

- 2) *Cluster Separation:* ( $C_{\text{sep}}$ ) evaluates the distinctness between different clusters by measuring the distances between their weighted centroids. For a set of clusters  $C$ , it is defined as

$$C_{\text{sep}}(C) = \sum_{c \in C} \sum_{d \in C, d \neq c} \text{dist}(\mu_c, \mu_d) \quad (13)$$

where  $\mu_c$  and  $\mu_d$  are the centroids of clusters  $c$  and  $d$ , and “dist” is the Euclidean distance. Higher separation values indicate more distinct and well-defined clusters. A core feature of our framework is the ability to self-tune the optimal number of clusters,  $K$ , using these metrics. The update rule for  $K$  is dynamically guided by an RL agent

$$K_{\text{new}} = K_{\text{cnt}} + \alpha_{\text{adp}} \times (C_{\text{comp}}(C) - \beta_{\text{adp}} \times C_{\text{sep}}(C)) \quad (14)$$

where  $K_{\text{new}}$  represents the new number of clusters and  $K_{\text{cnt}}$  is the current number of clusters. In addition, the adaptive tuning parameters,  $\alpha_{\text{adp}}$  and  $\beta_{\text{adp}}$ , are learned by the RL agent, which jointly optimizes cluster granularity and participant selection. The detailed formulation of this agent is presented in Section IV-C.

Finally, the entire process operates under key operational constraints, most notably a bounded communication delay, ensuring that the time delay ( $\tau_{\text{ie}}$ ) for any update remains within a predefined maximum,  $\Delta_{\text{ie}}$ , to maintain system responsiveness. The clustering process is guided by several key metrics designed to evaluate the quality of the cluster formations. The foundational metric is the  $d_W(i, j)$ , which quantifies the proximity between a device  $i$  and a cluster's center. The center of each cluster  $j$  is determined by its  $\mu_j$ , computed as the average position of devices weighted by their importance,  $\mu_j = \sum_{i \in c_j} W_i D_i / \sum_{i \in c_j} W_i$ , where  $D_i$  is the feature vector for device  $i$ . Crucially, the importance weight  $W_i$  is defined by the device's reputation score  $R_i$ , ensuring that more trustworthy devices have a greater influence on the cluster's center. Based on this, we define two quality metrics:  $C_{\text{comp}}$ , which measures the internal cohesion of a cluster, and  $C_{\text{sep}}$ , which evaluates the distinctness between different clusters.

Algorithm 1 presents the pseudocode of the proposed weighted adaptive clustering algorithm (WACA), designed to ensure robust convergence under stringent computational, energy, and communication constraints of IoT devices. By enforcing the bounded local update condition  $\|\Delta w_{\text{it}}\| \leq B_r$ ,

**Algorithm 1** Proposed WACA

---

**Input:**  $\mathcal{D}$ : IoT devices,  $K_{\min}, K_{\max}$   
**Output:** Final clusters  $\mathcal{C}$

- 1  $K \leftarrow \text{INITK}(K_{\min}, K_{\max})$ , initialize  $\{\mu_j\}_{j=1}^K$ ,  
 $\text{conv} \leftarrow \text{false}$
- 2 **while** not conv **do**
- 3    $\Delta \leftarrow 0$
- 4   **for**  $d_i \in \mathcal{D}$  **do**
- 5      $c_j^* \leftarrow \arg \min_{c_j} d_W(d_i, \mu_j)$
- 6     **if** assignment changed **then**
- 7        $\text{update } d_i \rightarrow c_j^*, \Delta += 1$
- 8    $\text{update } \mu_j = \frac{\sum_{i \in c_j} W_i D_i}{\sum_{i \in c_j} W_i}, \forall j$
- 9   **if**  $\Delta \leq \epsilon_{\text{conv}} |\mathcal{D}|$  **then**
- 10      $\text{conv} \leftarrow \text{true}$
- 11   **if** iteration mod  $T_{\text{adapt}} = 0$  **then**
- 12      $\mathcal{C}_{\text{comp}} \leftarrow \text{COMP}(\mathcal{C}), \mathcal{C}_{\text{sep}} \leftarrow \text{SEP}(\mathcal{C})$
- 13      $K' \leftarrow K + \alpha_{\text{adp}}(\mathcal{C}_{\text{comp}} - \beta_{\text{adp}} \mathcal{C}_{\text{sep}})$
- 14      $K \leftarrow \text{CLIP}(K', K_{\min}, K_{\max})$
- 15     **if**  $K$  changed **then**
- 16        $\text{reinitialize } \mathcal{C}$
- 17 **return**  $\mathcal{C}$

---

where  $B_i$  reflects device-specific resource limits, WACA stabilizes local learning, mitigates divergence during clustering, and achieves smooth, consistent convergence across heterogeneous IoT environments.

*Proof of Convergence for WACA:* The objective function monotonically decreases and is bounded below, ensuring convergence. The convergence analysis of the WACA follows a standard two-step iterative optimization framework analogous to classical Lloyd's algorithm, but extended to incorporate device-specific weights  $W_i$  that reflect heterogeneous computation and communication capabilities. The proof demonstrates that, at each iteration, both the assignment and centroid update steps do not increase the global objective function  $L(C)$ , thereby guaranteeing convergence to a local optimum.

- 1) *Step 1: Assignment:* At iteration  $t$ , each device  $D_i$  is assigned to the cluster with the nearest centroid

$$c_j^{t+1} = \left\{ i : j = \arg \min_l \|D_i - C_l^t\|^2 \right\}. \quad (15)$$

This ensures

$$\sum_{j, i \in c_j^{t+1}} W_i \|D_i - C_j^t\|^2 \leq L(C^t). \quad (16)$$

In this step, each device seeks the most suitable cluster centroid based on the minimum weighted Euclidean distance, guaranteeing a nonincreasing behavior of the loss function. Since the device weights  $W_i$  are positive and fixed during each iteration, the reassignment process strictly improves or maintains the objective value.

- 2) *Step 2: Centroid Update:* Each centroid is updated as the weighted mean

$$C_j^{t+1} = \frac{\sum_{i \in c_j^{t+1}} W_i D_i}{\sum_{i \in c_j^{t+1}} W_i} \quad (17)$$

which minimizes the weighted intracluster error

$$\sum_{i \in c_j^{t+1}} W_i \|D_i - C_j^{t+1}\|^2 \leq \sum_{i \in c_j^{t+1}} W_i \|D_i - C_j^t\|^2. \quad (18)$$

This update step ensures that, for fixed cluster assignments, the new centroid location achieves the minimal possible intracluster variance. Thus, each iteration yields an updated set of centroids that either reduces or maintains the current objective function value.

Combining the assignment and update steps gives

$$L(C^{t+1}) \leq L(C^t). \quad (19)$$

Since  $L(C) \geq 0$  and the sequence  $\{L(C^t)\}$  is nonincreasing and bounded, it converges. Convergence occurs when

$$L(C^{t+1}) = L(C^t) \quad (20)$$

at which point assignments and centroids stabilize. Therefore, the iterative process of WACA guarantees convergence to a locally optimal partition of the IoT devices, reflecting stable clustering behavior under heterogeneous resource conditions.

3) *RL-Driven Participant Selection (RL-AUDPS):* To efficiently handle heterogeneous IoT devices, RL-AUDPS selects a subset of devices per cluster based on a utility score

$$U_i = w_C \frac{C_i}{C_{\max}} + w_B \frac{B_i}{B_{\max}} + w_Q \frac{Q_i}{Q_{\max}} + w_R \frac{R_i}{R_{\max}} \quad (21)$$

where  $C_i$ ,  $B_i$ ,  $Q_i$ , and  $R_i$  denote the computation, battery, data quality, and reputation, respectively, and the weights satisfy

$$w_C + w_B + w_Q + w_R = 1. \quad (22)$$

Devices with higher  $U_i$  are preferentially selected, balancing resources, trust, and data quality. The weights ( $w_C, w_B, w_Q, w_R$ ) are dynamically adapted to global objectives like accuracy and energy efficiency. This adaptive weighting strategy allows the RL agent to learn optimal trade-offs among competing objectives under dynamically changing IoT network conditions. The RL-based decision process enables context-aware participant selection, where the policy  $\pi_{\text{RL}}$  continuously refines its selection strategy by incorporating feedback in the form of rewards derived from model convergence, latency, and communication efficiency metrics. Algorithm 2 provides the pseudocode of RL-AUDPS, which details the interaction between clustering, participant selection, and policy adaptation during training.

4) *Participant Selection and Adaptive Quota Management:* Once devices were clustered, the framework utilizes the RL-AUDPS mechanism to choose a subset of devices for each training round. The selection probability  $P_i$  for each device is proportional to its dynamically calculated composite utility score ( $U_i$ ), ensuring that more capable and reliable devices are favored. The number of participants selected from each cluster, denoted as  $n_s$ , is dynamically adapted to balance

**Algorithm 2** RL-AUDPS Mechanism

---

**Input:**  $\mathcal{D}$ : Set of IoT devices;  $T_{global}$ : Total communication rounds

**Output:**  $\mathbf{w}_{global}, \pi_{RL}, \mathbf{r}_{all}$

```

1  $\mathbf{w}_{global} \leftarrow \text{INITIALIZEMODEL}()$ 
2  $\mathbf{r}_{all}[i] \leftarrow 0.5$ 
3 Initialize RL policy  $\pi_{RL}$ 
4 for  $t = 1$  to  $T_{global}$  do
5    $S_t \leftarrow \text{OBSERVE SYSTEM STATE}(\mathbf{w}_{global}, \mathbf{r}_{all}, \text{metrics})$ 
6    $A_t \leftarrow \pi_{RL}(S_t)$ 
7    $\mathcal{C} \leftarrow \text{ADAPTIVE CLUSTERING}(\mathcal{D}, K)$ 
8    $\mathcal{D}_{selected} \leftarrow \emptyset$ 
9   for  $c$  in  $\mathcal{C}$  do
10    for  $d_i$  in  $c$  do
11      Compute utility score  $U_i$ 
12    end Select top  $\mathcal{D}_{c,sel} \subseteq c$  based on  $U_i$ 
13     $\mathcal{D}_{selected} \leftarrow \mathcal{D}_{selected} \cup \mathcal{D}_{c,sel}$ 
14  end
15  LocalUpdates  $\leftarrow \emptyset$ 
16  for  $d_i$  in  $\mathcal{D}_{selected}$  do
17     $\Delta \mathbf{w}_i \leftarrow \text{LOCAL TRAIN}(d_i, \mathbf{w}_{global})$ 
18    Add  $\Delta \mathbf{w}_i$  to LocalUpdates
19  end
20   $\Delta \mathbf{w}_{agg} \leftarrow \text{AGGREGATE}(\text{LocalUpdates}, \mathbf{r}_{all})$ 
21   $\mathbf{w}_{global} \leftarrow \mathbf{w}_{global} + \eta_{global} \cdot \Delta \mathbf{w}_{agg}$ 
22  Update  $\mathbf{r}_{all}$  based on contribution quality
23  Compute reward  $r_t$  for action  $A_t$  in state  $S_t$ 
24  Update policy  $\pi_{RL}$  using  $(S_t, A_t, r_t)$ 
25 end
26 return  $\mathbf{w}_{global}, \pi_{RL}, \mathbf{r}_{all}$ 

```

---

participation diversity with communication overhead. This selection is calculated as follows:

$$n_s = \left\lfloor \rho_s \times N_c \times \left( 1 + \delta_a \times \frac{R_{\text{available}}}{R_{\text{max}}} \right) \right\rfloor \quad (23)$$

where  $\rho_s$  is the base selection ratio, representing the default proportion of clients to be selected per cluster, while  $N_c$  is the number of available clients in the cluster. The factor  $\delta_a$  is the adaptive scaling coefficient that adjusts selection based on current system conditions.  $R_{\text{available}}$  denotes the available communication or computational resources (e.g., bandwidth or processing power), and  $R_{\text{max}}$  is the maximum reference capacity of these resources. Both  $\rho_s$  and  $\delta_a$  are dynamically determined in each round by the policy of an RL agent, enabling adaptive control over client selection to balance participation diversity and resource utilization.

5) *RL Agent Formulation*: To dynamically optimize the participant selection process, we formulate the problem as a Markov decision process (MDP), which is solved using a Q-learning agent. The agent's goal is to learn an optimal policy,  $\pi : S \rightarrow A$ , that maximizes the cumulative discounted reward. The core components of this MDP are defined as follows.

a) *State space (S)*: The state  $S_t$  at time  $t$  provides a snapshot of the global system's performance. It is a vector composed of key metrics from the previous training round, including global model accuracy and its rate of change, average device energy consumption, overall communication overhead, and metrics for cluster quality, such as compactness and separation.

b) *Action space (A)*: The action  $A_t$  selected by the agent is a vector of parameters that controls the entire selection process for the subsequent round. This action vector jointly determines all adaptive parameters used in (21) and (23)

$$A_t = [w_C, w_B, w_Q, w_R, \rho_s, \delta_a]. \quad (24)$$

Here,  $w_C, \dots, w_R$  are the utility weights used in (21), and  $\rho_s$  and  $\delta_a$  are the parameters for adaptive quota management from (23).

Given the complexity of the 6-D action space, a uniform discretization would be computationally intractable. Therefore, to make the learning process feasible, we defined a discrete set of high-level strategic policies. Instead, we defined a discrete set of high-level strategic policies. For example, these policies include accuracy-focused mode (prioritizing devices with high data quality and reputation), energy-saving mode (prioritizing devices with high battery life), and balanced mode. The agent's task is simplified to selecting the most suitable strategic policy from this predefined discrete set in response to the current system state  $S_t$ .

c) *Reward function ( $R_t$ )*: The reward function  $R_t$  guides the agent's learning by providing a scalar feedback signal after each round. It is explicitly designed to balance the trade-off between improving model performance and minimizing resource consumption

$$R_t = ((\lambda_{\text{acc}} \times A_t) - (\lambda_{\text{en}} \times E_t) - (\lambda_{\text{comm}} \times C_t)) \quad (25)$$

where  $A_t$  is the improvement in global model accuracy, while  $E_t$  and  $C_t$  are the total energy and communication costs incurred in the round, respectively. The coefficients  $\lambda_{\text{acc}}$ ,  $\lambda_{\text{en}}$ , and  $\lambda_{\text{comm}}$  are the hyperparameters that balance the importance of these competing objectives. By learning to maximize this cumulative reward, the agent develops a sophisticated policy for efficient and robust FL.

6) *TA-RMSProp Optimization*: Standard RMSProp optimization uses a fixed decay factor ( $\gamma$ ) for all devices, ignoring their varying reliability. Our proposed trust-adaptive RMSProp (TA-RMSProp) enhances this by making the decay factor,  $\gamma_i$ , dynamic and dependent on each device's reputation score,  $R_i$ . This personalizes the optimization process as follows.

- 1) *High-Reputation Devices ( $R_i \rightarrow 1$ )*: These reliable devices are assigned a higher decay factor ( $\gamma_i$ ). This gives more weight to their historical gradients, leading to a stable and conservative learning rate that trusts their consistent performance.
- 2) *Low-Reputation Devices ( $R_i \rightarrow 0$ )*: These unreliable devices receive a lower decay factor ( $\gamma_i$ ). This reduces the influence of their noisy past gradients and makes the optimizer more reactive to their most recent update, allowing it to adapt quickly to erratic behavior.



The adaptive decay factor  $\gamma_i$  for each device  $i$  is calculated as a linear function of its reputation score

$$\gamma_i = \gamma_{\text{base}} + ((\gamma_{\text{max}} - \gamma_{\text{base}}) \times R_i) \quad (26)$$

where  $\gamma_{\text{base}}$  and  $\gamma_{\text{max}}$  are the predefined minimum and maximum bounds for the decay factor (e.g., 0.8 and 0.99). This adaptive  $\gamma_i$  is then used in the RMSProp moving average update rule for each device

$$E[g_t^2]_i = (\gamma_i \times E[g_{t-1}^2]_i + (1 - \gamma_i) \times g_t^2). \quad (27)$$

*a) Convergence and trust-aware behavior analysis:*

The TA-RMSProp update retains the convergence properties of standard RMSProp, as  $\gamma_i \in [\gamma_{\text{base}}, \gamma_{\text{max}}] \subset (0, 1)$ . High-reputation devices contribute stable gradient history, ensuring smooth convergence, while low-reputation devices adapt quickly to new gradients, preventing erratic updates from destabilizing the global model. Formally, the expected squared gradient moving average is bounded

$$(1 - \gamma_{\text{max}}) g_{\min}^2 \leq E[g_t^2]_i \leq g_{\max}^2 \quad (28)$$

guaranteeing that the per-device learning rate  $\eta/(E[g_t^2]_i + \epsilon)^{1/2}$  remains finite and stable. This ensures that, under heterogeneous trust conditions, TA-RMSProp converges to a stationary point while dynamically weighting contributions according to device reliability.

This synergistic design directly links high-level trust management to low-level optimization, improving robustness and efficiency in heterogeneous, resource-constrained IoT environments.

*b) Integration with reputation system:* The reputation score  $R_i$  for each device is continuously updated based on the consistency and reliability of its local updates. Aggregation across multiple coordinators ensures that the global reputation reflects both individual and collective behavior. TA-RMSProp leverages these scores to weight device contributions adaptively, limiting the impact of potentially malicious or noisy updates. In addition, devices with previously low reputation can gradually recover as their performance stabilizes, providing resilience against transient failures and mitigating manipulation attempts. This integration ensures that the optimization process remains robust, stable, and fair over long-term deployments in heterogeneous IoT environments.

*7) SCH Selection and Intercluster Gossip Protocol:* Each cluster dynamically selects a representative node, the SCH, to manage local aggregation and communicate with the coordination level. The SCH is chosen as the device with the highest utility score ( $U_i$ ), determined by the RL-AUDPS mechanism in (24). This ensures that the most capable node, balancing computational resources, energy, data quality, and reputation, is selected for this critical role. Once selected, these SCHs use an enhanced gossip protocol to exchange aggregated model information between clusters. The protocol's efficiency and resilience are enhanced through two key adaptive features as follows.

- 1) *Trust-Aware Peer Selection (TAPS):* This replaces random peer selection in conventional gossip protocols with a trust-based approach. Each SCH selects peers with

high reputation scores, reflecting historical reliability and contribution quality, to ensure efficient update propagation while minimizing the influence of unreliable nodes.

- 2) *Adaptive Gossip Frequency (AGF):* This mechanism dynamically adjusts the rate of gossip exchanges—reducing communication during network congestion or slow convergence to save resources, and increasing it when the model is learning rapidly to accelerate convergence. This balances communication costs with learning efficiency.

To strengthen the security and reliability of intercluster communication, we adopt the TAGS, as detailed in Algorithm 3. Each SCH selects a subset of peers with the highest reputation scores (TAPS), propagates updates, validates received messages, and performs a weighted aggregation based on peer trustworthiness. The AGF dynamically adjusts propagation rounds according to convergence speed and network conditions. This design ensures the following.

- 1) Resilience to up to  $f$  Byzantine nodes per cluster via local validation and weighted aggregation.
- 2) Resistance to adversarial manipulation, as malicious updates from low-reputation nodes have limited influence.

---

**Algorithm 3** Trust-Aware Gossip Protocol (TAGS) for Inter-cluster Communication

---

**Input:** SCH set  $\mathcal{S}$ , local updates  $\Delta \mathbf{w}_{SCH}$ , reputation scores  $R_i$ , maximum peers  $P_{\text{max}}$ , max rounds  $T_g$

**Output:** Refined aggregated update  $\Delta \mathbf{w}_{agg}$

```

1 Initialize  $\Delta \mathbf{w}_{agg} \leftarrow 0$ ,  $total\_weight \leftarrow 0$ 
2 for  $t = 1$  to  $T_g$  do
3   foreach  $SCH_i \in \mathcal{S}$  do
4     Select  $\mathcal{C}_i = \{SCH_1, \dots, SCH_P\}$  with highest  $R_j$ 
5     Send  $\Delta \mathbf{w}_{SCH_i}$  to  $\mathcal{C}_i$ 
6     Receive  $\Delta \mathbf{w}_{peer}$  from  $\mathcal{C}_i$ 
7     Validate  $\Delta \mathbf{w}_{peer}$  using statistical consistency and norm checks
8      $\Delta \mathbf{w}_{agg} \leftarrow \Delta \mathbf{w}_{agg} + \sum_{i \in \mathcal{C}_i} R_i \cdot \Delta \mathbf{w}_{peer}$ 
9      $total\_weight \leftarrow total\_weight + \sum_{i \in \mathcal{C}_i} R_i$ 
10  end
11 end
12 Normalize  $\Delta \mathbf{w}_{agg}$  using  $total\_weight$  if needed

```

---

- 3) Efficient, bounded communication overhead suitable for large-scale IoT deployments.

*B. Coordination Level*

The coordination level acts as an intermediary layer between the edge clusters and the global aggregator, enhancing both scalability and robustness. Its primary functions are to aggregate model updates from the edge and facilitate efficient communication among coordinators.

First, each coordinator receives model updates from the SCH it supervises. The SCH, being the most trustworthy

device in its cluster, performs preliminary lightweight validation on these updates, such as checking their statistical properties. This proactive filtering helps mitigate the spread of corrupted or malicious data early in the hierarchy.

Second, to maintain system-wide consistency, the coordinators communicate among themselves using an enhanced, TAGS.

---

**Algorithm 4** Coordination Level Mechanism
 

---

**Input:** Set of SCHs  $\mathcal{S}$ , updates  $\Delta \mathbf{w}_{s_i}$ , learning rate  $\eta$   
**Output:** Updated global model  $\mathbf{w}_{global}$

```

1  $\Delta \mathbf{w}_{agg} \leftarrow 0$ ,  $total\_weight \leftarrow 0$ 
2 for each  $s_i \in \mathcal{S}$  do
3    $w_i \leftarrow$ 
     DYNAMICWEIGHTINGFUNCTION( $s_i, timestamp_i$ )
4    $\Delta \mathbf{w}_{agg} \leftarrow \Delta \mathbf{w}_{agg} + w_i \cdot \Delta \mathbf{w}_{s_i}$ 
5    $total\_weight \leftarrow total\_weight + w_i$ 
6 end
7 if  $total\_weight > 0$  then
8    $\Delta \mathbf{w}_{agg} \leftarrow \Delta \mathbf{w}_{agg} / total\_weight$ 
9 end
10  $\mathbf{w}_{global}^{new} \leftarrow \mathbf{w}_{global} + \eta \cdot \Delta \mathbf{w}_{agg}$ 
11 if  $\|\mathbf{w}_{global}^{new} - \mathbf{w}_{global}\| < \varepsilon$  then
12    $\mathbf{w}_{global} \leftarrow \mathbf{w}_{global}^{new}$ 
13   break
14 end
15  $\mathbf{w}_{global} \leftarrow \mathbf{w}_{global}^{new}$ 
16 for each  $s_i \in \mathcal{S}$  do
17   DISTRIBUTEModelToSCH( $s_i, \mathbf{w}_{global}$ )
18 end
19 return  $\mathbf{w}_{global}$ 

```

---

This intercoordinator refinement process produces a more robust aggregated model to be sent to the global aggregator, as detailed in Algorithm 4.

*Operational Constraints and Role of the SCH:* The coordination level operates under constrained network and resource conditions. Communication between each edge cluster (via its SCH) and the coordinator node is limited by the maximum available bandwidth  $\Delta_{ec}$  to avoid congestion. Although coordinators have higher capacity, they remain bound by computation, memory, and energy limits to ensure reliable operation.

Within each cluster  $h$ , the super cluster head (SCH) is defined as

$$SCH = \arg \max_{i \in \mathcal{E}_h} U_i \quad (29)$$

where  $U_i$  denotes the utility score of client  $i$ . The selected SCH aggregates validated local model updates  $\Delta \mathbf{w}_{h,e}$  from all members  $e \in \mathcal{E}_h$  to form a cluster-level update

$$\Delta \mathbf{w}_{SCH} = \frac{1}{|\mathcal{E}_h|} \sum_{e \in \mathcal{E}_h} \Delta \mathbf{w}_{h,e}. \quad (30)$$

As the most capable client, the SCH performs lightweight validation (e.g., update-norm and range checks) before aggregation to filter corrupted submissions. It then forwards the

refined update to its coordinator and disseminates the updated global model back to cluster members. This local validation serves as a preliminary defense, while comprehensive verification and reputation management are handled by the global aggregator, as detailed in Section IV-C.

*1) Dynamic Trust-Aware Communication and Delay Mitigation:* To maintain global consistency and efficient knowledge dissemination, coordinators adopt a dynamic communication protocol combining TAPS and AGF: TAPS selects reliable peers, while AGF optimizes timing based on network and learning dynamics, enabling a self-regulating system for improved convergence and communication efficiency.

To address edge cluster delays, coordinators apply a three-tier aggregation strategy: 1) a buffer to store asynchronous updates; 2) delay-tolerant aggregation without waiting for all clusters; and 3) delay-aware weighting, where each update's contribution  $\lambda_e$  is adjusted based on its arrival delay  $\tau_e$ , favoring fresher information.

To handle arriving updates asynchronously and prevent data loss from timing mismatches, each coordinator node implements an aggregation buffer. The state of the buffer at coordinator  $c$  at time  $t$ , denoted as  $B_c(t)$ , is formally defined as  $B_c(t) = \{\Delta \mathbf{w}_e^{t'} \mid e \in \mathcal{E}, t - \tau_{\text{valid}} \leq t' \leq t\}$ , where  $\tau_{\text{valid}}$  represents the maximum allowable delay for an update to be considered valid for the current aggregation cycle. This buffering mechanism ensures that updates are not discarded due to network latency.

### C. Global Aggregator Level

The global aggregator is the final and highest tier in the proposed hierarchical architecture. Its primary responsibility is to synthesize the model updates received from all coordinator nodes to produce the final, converged global model. This level performs two critical functions to ensure the long-term integrity and stability of the system: 1) it employs on-the-fly anomaly detection to identify and filter out statistical outliers or potentially malicious updates from coordinators and 2) it performs a reputation-based weighted aggregation of the valid updates, giving more influence to coordinators with a history of reliable contributions. This final, secure aggregation step is crucial for maintaining the performance and reliability of the entire FL process. The pseudocode for the global aggregator is given in Algorithm 5.

*1) Coordinator Reputation Update Mechanism:* Similar to the reputation system at the edge level, the global aggregator maintains a dynamic reputation score ( $R_c$ ) for each coordinator  $c$ . This score is crucial for the reputation-based weighting performed during the final aggregation step. A coordinator's reputation is updated after each global communication round based on the quality and consistency of the aggregated update it provides.

The quality of a coordinator's contribution is determined by the on-the-fly anomaly detection mechanism. Specifically, we define a coordinator contribution score ( $CCS_c^t$ ) for coordinator  $c$  in round  $t$ , which is based on the similarity of its submitted update to the network consensus

$$CCS_c^t = \text{sim}(\Delta \mathbf{w}_c^t, \Delta \bar{\mathbf{w}}^t) \quad (31)$$

**Algorithm 5** Global Aggregation With Dynamic Weighting

---

**Input:** Set of SCHs  $\mathcal{S}$ , updates  $\Delta \mathbf{w}_{s_i}$ , learning rate  $\eta$   
**Output:** Updated global model  $\mathbf{w}_{global}$

```

1  $\Delta \mathbf{w}_{agg} \leftarrow 0, total\_weight \leftarrow 0$ 
2 for each  $s_i \in \mathcal{S}$  do
3    $w_i \leftarrow$ 
     DYNAMICWEIGHTINGFUNCTION( $s_i, timestamp_i$ )
4    $\Delta \mathbf{w}_{agg} \leftarrow \Delta \mathbf{w}_{agg} + w_i \cdot \Delta \mathbf{w}_{s_i}$ 
5    $total\_weight \leftarrow total\_weight + w_i$ 
6 end
7 if  $total\_weight > 0$  then
8    $\Delta \mathbf{w}_{agg} \leftarrow \Delta \mathbf{w}_{agg} / total\_weight$ 
9 end
10  $\mathbf{w}_{global}^{new} \leftarrow \mathbf{w}_{global} + \eta \cdot \Delta \mathbf{w}_{agg}$ 
11 if  $\|\mathbf{w}_{global}^{new} - \mathbf{w}_{global}\| < \varepsilon$  then
12    $\mathbf{w}_{global} \leftarrow \mathbf{w}_{global}^{new}$ 
13   break
14 end
15  $\mathbf{w}_{global} \leftarrow \mathbf{w}_{global}^{new}$ 
16 for each  $s_i \in \mathcal{S}$  do
17   DISTRIBUTEModelToSCH( $s_i, \mathbf{w}_{global}$ )
18 end
19 return  $\mathbf{w}_{global}$ 

```

---

where  $\Delta \mathbf{w}_c^t$  is the update from coordinator  $c$ ,  $\Delta \bar{\mathbf{w}}^t$  is the mean of all valid updates received in that round, and  $\text{sim}(\cdot)$  is the normalized cosine similarity.

A contribution is considered positive if its similarity score is above the predefined anomaly threshold ( $\theta_{anom}$ ). The reputation score  $R_c$  is then updated using the same growth ( $\gamma_r$ ) and decay ( $\delta_r$ ) factors used for devices, ensuring a consistent trust model across the framework.

- 1) For a positive contribution ( $\text{CCS}_c^t \geq \theta_{anom}$ ), the coordinator's reputation increases, pushing it toward 1

$$R_c^{t+1} = R_c^t + (\gamma_r \times (1 - R_c^t)). \quad (32)$$

- 2) For an anomalous contribution ( $\text{CCS}_c^t < \theta_{anom}$ ), the coordinator's reputation is penalized, pushing it toward 0

$$R_c^{t+1} = R_c^t - (\delta_r \times R_c^t). \quad (33)$$

This explicit update rule ensures that the reputation-based weighting at the global level is founded on a transparent, dynamic, and robust trust assessment of the coordinators.

#### D. Global Model Convergence Analysis

We analyze the convergence behavior of the global model sequence  $\{\mathbf{w}^t\}$  generated by the proposed RL-UDHFL framework. The analysis follows standard assumptions in distributed optimization.

- A1 (*Strong Convexity*):  $F(\mathbf{w})$  is  $\mu$ -strongly convex.
- A2 (*L-Smoothness*): Each  $F_k(\mathbf{w})$  has  $L$ -Lipschitz continuous gradients.
- A3 (*Bounded Variance*):  $\mathbb{E}[\|\nabla F_k(\mathbf{w}) - \nabla F(\mathbf{w})\|^2] \leq \sigma^2$ .

- A4 (*Bounded Local Updates*):  $\|\Delta \mathbf{w}_k\| \leq G$ , implying  $\|\Delta \mathbf{w}_{global}^t\| \leq G_{global}$ .

We now show that the hierarchical aggregation preserves descent and guarantees convergence.

1) *Bounded Update Propagation Across Levels*: Since the global update is a weighted sum of client-level updates across cluster and coordination layers, assumption A4 implies

$$\|\Delta \mathbf{w}_{global}^t\| \leq \sum_{h,c,k} \alpha_{h,c,k} \|\Delta \mathbf{w}_k^t\| \leq G_{global} \quad (34)$$

where  $\alpha_{h,c,k}$  is normalized hierarchical weights.

2) *Descent Under L-Smoothness*: Applying the descent lemma

$$F(\mathbf{w}^{t+1}) \leq F(\mathbf{w}^t) - \eta \langle \nabla F(\mathbf{w}^t), \Delta \mathbf{w}_{global}^t \rangle + \frac{L\eta^2}{2} \|\Delta \mathbf{w}_{global}^t\|^2. \quad (35)$$

Taking expectation and substituting boundedness

$$\mathbb{E}[F(\mathbf{w}^{t+1})] \leq F(\mathbf{w}^t) - \eta \langle \nabla F(\mathbf{w}^t), \mathbb{E}[\Delta \mathbf{w}_{global}^t] \rangle + \frac{L\eta^2 G_{global}^2}{2}. \quad (36)$$

3) *Gradient Approximation Error*: Let the hierarchical gradient bias be bounded as

$$\|\mathbb{E}[\Delta \mathbf{w}_{global}^t] - \nabla F(\mathbf{w}^t)\| \leq \epsilon \quad (37)$$

which captures effects of multilevel aggregation and device heterogeneity.

Using (37)

$$\langle \nabla F(\mathbf{w}^t), \mathbb{E}[\Delta \mathbf{w}_{global}^t] \rangle \geq \|\nabla F(\mathbf{w}^t)\|^2 - \epsilon \|\nabla F(\mathbf{w}^t)\|. \quad (38)$$

Substituting back into (36) yields

$$\mathbb{E}[F(\mathbf{w}^{t+1})] \leq F(\mathbf{w}^t) - \eta \|\nabla F(\mathbf{w}^t)\|^2 + \eta \epsilon \|\nabla F(\mathbf{w}^t)\| + \frac{L\eta^2 G_{global}^2}{2}. \quad (39)$$

4) *Convergence to Stationary Point*: For  $\eta < 1/L$  and sufficiently small gradient bias  $\epsilon$ , the negative quadratic term dominates

$$\|\nabla F(\mathbf{w}^t)\|^2 \geq \epsilon \|\nabla F(\mathbf{w}^t)\| + O(\eta). \quad (40)$$

Summing (39) over  $t = 1$  to  $T$  and using telescoping yields

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E} \|\nabla F(\mathbf{w}^t)\|^2 \leq O(\eta) + O(\epsilon). \quad (41)$$

Thus,

$$\lim_{t \rightarrow \infty} \mathbb{E} \|\nabla F(\mathbf{w}^t)\| = 0 \quad (42)$$

proving convergence to a stationary point of  $F(\mathbf{w})$ .

## V. CONVERGENCE ANALYSIS OF RL-UDHFL

In this section, we present a rigorous mathematical proof of convergence for the proposed RL-UDHFL framework. The analysis encompasses device-level updates, cluster-level aggregation, coordinator-level aggregation, and global model convergence under bounded gradients, Lipschitz smoothness, and trust-weighted aggregation, establishing convergence to a stationary point

$$\begin{aligned}
\gamma_k &= \gamma_{\text{base}} + (\gamma_{\text{max}} - \gamma_{\text{base}}) R_k \\
E[g_k^2]^t &= \gamma_k E[g_k^2]^{t-1} + (1 - \gamma_k) (g_k^t)^2 \\
\Delta \mathbf{w}_k^t &= -\eta \frac{g_k^t}{\sqrt{E[g_k^2]^t} + \epsilon} \\
\mathbf{w}_k^{t+1} &= \mathbf{w}_k^t + \Delta \mathbf{w}_k^t \\
\Delta \mathbf{w}_h^t &= \frac{\sum_{k \in \mathcal{E}_h} \alpha_k \Delta \mathbf{w}_k^t}{\sum_{k \in \mathcal{E}_h} \alpha_k}, \quad \|\Delta \mathbf{w}_h^t\| \leq G_h \\
\Delta \mathbf{w}_{\text{coord}}^t &= \frac{\sum_{h=1}^H \beta_h \Delta \mathbf{w}_h^t}{\sum_{h=1}^H \beta_h}, \quad \|\Delta \mathbf{w}_{\text{coord}}^t\| \leq G_{\text{coord}} \\
\mathbf{w}^{t+1} &= \mathbf{w}^t - \eta \Delta \mathbf{w}_{\text{coord}}^t \\
&= \mathbf{w}^t - \eta \sum_{k=1}^N \lambda_k^t \Delta \mathbf{w}_k^{t-\tau_k^t}, \quad \sum_{k=1}^N \lambda_k^t = 1 \\
F(\mathbf{w}^{t+1}) &\leq F(\mathbf{w}^t) - \eta \langle \nabla F(\mathbf{w}^t), \Delta \mathbf{w}_{\text{coord}}^t \rangle \\
&\quad + \frac{L\eta^2}{2} \|\Delta \mathbf{w}_{\text{coord}}^t\|^2 \\
\mathbb{E}[F(\mathbf{w}^{t+1})] &\leq F(\mathbf{w}^t) - \eta \langle \nabla F(\mathbf{w}^t), \mathbb{E}[\Delta \mathbf{w}_{\text{coord}}^t] \rangle \\
&\quad + \frac{L\eta^2}{2} \mathbb{E}[\|\Delta \mathbf{w}_{\text{coord}}^t\|^2] \\
\mathbb{E}[\Delta \mathbf{w}_{\text{coord}}^t] &\approx \nabla F(\mathbf{w}^t), \quad \|\Delta \mathbf{w}_{\text{coord}}^t\| \leq G_{\text{coord}} \\
\mathbb{E}[F(\mathbf{w}^{t+1})] &\leq F(\mathbf{w}^t) - \eta \|\nabla F(\mathbf{w}^t)\|^2 + \frac{L\eta^2 G_{\text{coord}}^2}{2} \\
\sum_{t=0}^{T-1} \eta \|\nabla F(\mathbf{w}^t)\|^2 &\leq F(\mathbf{w}^0) - F^* + \frac{TL\eta^2 G_{\text{coord}}^2}{2} \\
\min_t \|\nabla F(\mathbf{w}^t)\|^2 &\leq \frac{F(\mathbf{w}^0) - F^*}{\eta T} + \frac{L\eta G_{\text{coord}}^2}{2} \\
0 &\leq \lim_{t \rightarrow \infty} \mathbb{E}[\|\nabla F(\mathbf{w}^t)\|^2] \rightarrow 0.
\end{aligned}$$

The above derivation demonstrates that RL-UDHFL converges to a stationary point under bounded gradient assumptions and Lipschitz smoothness, confirming theoretical soundness for practical IoT deployment.

### A. Scalability Analysis

This section presents a formal analysis of the scalability of the proposed hierarchical RL-UDHFL framework in terms of communication and energy efficiency as the number of IoT devices grows. Let  $N$  be the total number of devices and  $K$  the number of clusters, with each cluster containing on average  $\bar{n} = N/K$  devices.

1) *Communication Complexity*: In standard flat FL, each device sends a full model update of size  $d$  to the server, yielding a per-round communication cost

$$\text{Comm}_{\text{flat}} = O(Nd). \quad (43)$$

In the hierarchical design, communication occurs at the edge (intracluster aggregation) and coordination levels. The resulting communication per round is

$$\text{Comm}_{\text{hier}} = O(Kd) + O(\bar{n}d) = O\left(d\left(K + \frac{N}{K}\right)\right). \quad (44)$$

Minimizing  $K + N/K$  gives

$$\frac{\partial}{\partial K} \left(K + \frac{N}{K}\right) = 0 \Rightarrow K^* = \sqrt{N}. \quad (45)$$

Substituting  $K^*$

$$\text{Comm}_{\text{hier}}(K^*) = O(d\sqrt{N}) \quad (46)$$

showing that the communication complexity improves from linear  $O(N)$  to sublinear  $O((N)^{1/2})$  scaling.

2) *Energy Consumption Scaling*: Let  $E_t$  be the local training energy and  $E_c$  be the energy required for direct device-to-server communication. In flat FL, total energy is

$$E_{\text{flat}} = N(E_t + E_c). \quad (47)$$

In the hierarchical architecture, devices primarily perform low-power intracluster communication with energy cost  $E_c^{\text{edge}} \ll E_c$ , and only  $K$  cluster representatives perform higher-energy coordination-level communication. Thus, the energy consumption is

$$E_{\text{hier}} = N(E_t + E_c^{\text{edge}}) + KE_c^{\text{coord}}. \quad (48)$$

With  $K = \Theta((N)^{1/2})$ , this becomes

$$E_{\text{hier}} \approx NE_t + O(\sqrt{N}) \quad (49)$$

indicating significantly reduced communication-related energy expenditure compared to flat FL.

These results demonstrate that the hierarchical RL-UDHFL framework achieves sublinear communication scaling and improved energy efficiency. By selecting  $K = \Theta((N)^{1/2})$ , the overall system maintains efficient operation even in large-scale IoT deployments.

### B. Communication Cost and Energy Consumption With Utility-Driven Objective Analysis

This section presents a detailed analysis of communication cost, energy consumption, and their relationship with the utility-driven optimization objective of the proposed method.

1) *Communication Cost Analysis*: In the proposed method, communication occurs at two levels: intracluster aggregation and coordination. The communication cost for a given device  $i$  in a cluster  $\mathcal{C}_k$  is expressed as

$$\text{Comm}_i = O(d) \quad (50)$$

where  $d$  is the size of the model update. The total communication cost per round in the hierarchical system, considering  $K$  clusters and  $N$  devices, is given by

$$\text{Comm}_{\text{prop}} = O(K \cdot d) + O(\bar{n} \cdot d) \quad (51)$$

where  $K$  is the number of clusters, and  $\bar{n}$  is the average number of devices per cluster. To minimize communication cost, we



seek the optimal number of clusters  $K$ , which minimizes the total communication cost

$$K^* = \arg \min_K \left( O(K \cdot d) + O\left(\frac{N}{K} \cdot d\right) \right). \quad (52)$$

Solving the above optimization problem by differentiating with respect to  $K$  and setting the derivative equal to zero, we find

$$K^* = \sqrt{N}. \quad (53)$$

Substituting this value of  $K$  into the total communication cost, we obtain the final communication cost for the proposed method

$$\text{Comm}_{\text{prop}}(K^*) = O(d \cdot \sqrt{N}). \quad (54)$$

This demonstrates that the communication cost in the proposed system scales as  $O((N)^{1/2})$ , a significant reduction compared to flat FL methods.

2) *Energy Consumption Analysis*: The total energy consumption in the flat FL system is expressed as

$$E_{\text{flat}} = N(E_t + E_c) \quad (55)$$

where  $E_t$  is the local training energy, and  $E_c$  is the energy consumed in communication. In the proposed method, the energy consumption is distributed across two levels: edge-level communication and coordination-level communication. The total energy consumption per round in the hierarchical system is

$$E_{\text{prop}} = N(E_t + E_c^{\text{edge}}) + K \cdot E_c^{\text{coord}} \quad (56)$$

where  $E_c^{\text{edge}}$  is the energy used for intracluster communication, and  $E_c^{\text{coord}}$  is the energy used for coordination-level communication. To minimize energy consumption, we assume  $K = \Theta((N)^{1/2})$ , leading to the following simplification:

$$E_{\text{prop}} = NE_t + O(\sqrt{N}) \cdot E_c^{\text{coord}}. \quad (57)$$

This analysis shows that the hierarchical structure significantly reduces energy consumption compared to flat FL.

3) *Impact of Client Selection*: Client selection plays a critical role in balancing communication cost and energy consumption. The utility-driven objective function is formulated as

$$U(C_t) = \sum_{i \in C_t} (\alpha A_i - \beta E_i - \gamma L_i) \quad (58)$$

where  $A_i$ ,  $E_i$ , and  $L_i$  represent the expected accuracy, energy consumption, and communication cost of client  $i$ , respectively, and  $\alpha$ ,  $\beta$ , and  $\gamma$  are tunable parameters that balance these trade-offs. The total communication and energy costs for a selected set of clients  $C_t$  are given by

$$\text{Comm}_t = \sum_{i \in C_t} \text{Comm}_i, \quad E_{\text{total},t} = \sum_{i \in C_t} E_i. \quad (59)$$

4) *Optimization Parameters and RL Formulation*: The client selection process is guided by an RL agent, which seeks to maximize the reward function

$$\text{Reward}_t = -\alpha \cdot \text{Comm}_t - \beta \cdot E_{\text{total},t} + \gamma \cdot Q_{\text{avg},t} \quad (60)$$

where  $\text{Comm}_t$ ,  $E_{\text{total},t}$ , and  $Q_{\text{avg},t}$  represent the communication cost, total energy consumption, and average data quality at

time  $t$ , respectively. The RL agent adapts to the environment by selecting the optimal set of clients  $C_t$  based on these parameters, thereby optimizing communication, energy, and model accuracy in real-time.

5) *Analysis of the Trade-Offs Between Communication, Energy, and Accuracy*: The optimization of communication, energy consumption, and accuracy involves solving a multi-objective optimization problem. The balance between these objectives can be adjusted by tuning the weights  $\alpha$ ,  $\beta$ , and  $\gamma$ . The trade-off is analyzed by varying these weights and observing the changes in communication cost, energy consumption, and model accuracy over multiple rounds of training.

### C. Experimental Setup

To rigorously evaluate the performance of the proposed RL-UDHFL framework, we conducted simulations using four diverse datasets under an FL environment comprising 100 client devices. To emulate statistical heterogeneity commonly observed in real-world IoT networks, training data was partitioned across clients using a Dirichlet distribution with parameter  $\alpha = 0.5$ , resulting in a non-IID data setting.

- 1) *MNIST*: A standard dataset of handwritten digits, used to benchmark the framework in a controlled, low-complexity environment.
- 2) *Fashion-MNIST (FaMNIST)*: A drop-in replacement for MNIST that contains grayscale images of clothing items across ten classes. It offers moderate complexity, serving as a bridge between MNIST and more intricate datasets like CIFAR-10.
- 3) *CIFAR-10*: A more complex dataset comprising colored images across ten categories. It is employed to assess the framework's scalability and robustness under more challenging feature distributions.
- 4) *IoT-23*: A domain-specific, time-series dataset developed in-house to emulate a realistic IoT environment. It contains heterogeneous sensor data with simulated anomalies, enabling evaluation of the framework's performance in practical, anomaly-prone settings [38].

### D. Baseline Methods

We benchmark the proposed RL-UDHFL method against five representative FL algorithms to ensure a comprehensive performance comparison as follows.

- 1) *FedAvg* [25]: The foundational FedAvg algorithm that aggregates local models via simple mean.
- 2) *FedProx* [26]: An extension of FedAvg incorporating a proximal term to address statistical heterogeneity.
- 3) *SCAFFOLD* [38]: A variance-reduction method that utilizes control variates to mitigate client drift in non-IID settings.
- 4) *T-FedHA* [32]: A T-FedHA FL framework tailored for resource-constrained IoT deployments.
- 5) *HAFedRL* [36]: A hierarchical adaptive method that leverages RL for dynamic resource allocation.

### E. Model and Framework Configuration

For the MNIST, FaMNIST, and CIFAR-10 datasets, we employed convolutional neural network (CNN) models with

TABLE I  
FRAMEWORK HYPERPARAMETERS

| General Settings                                        |                           |
|---------------------------------------------------------|---------------------------|
| Communication Rounds                                    | 200                       |
| Clients                                                 | 100                       |
| Dirichlet $\alpha$                                      | 0.5                       |
| Local Epochs ( $E$ )                                    | 5                         |
| Batch Size ( $B$ )                                      | 32                        |
| Global Learning Rate ( $\eta$ )                         | 1.0                       |
| Local Learning Rate ( $\eta_{\text{local}}$ )           | 0.001                     |
| Optimizer                                               | TA-RMSProp ( $\rho=0.9$ ) |
| Adaptive Clustering                                     |                           |
| Cluster Range ( $K_{\min}, K_{\max}$ )                  | [5, 15]                   |
| RL Params ( $\alpha_{\text{adp}}, \beta_{\text{adp}}$ ) | 0.1, 0.5                  |
| RL Agent (RL-AUDPS)                                     |                           |
| Agent Type                                              | Q-learning                |
| RL LR / Discount ( $\alpha, \gamma$ )                   | 0.1 / 0.9                 |
| Utility Weights ( $w_C, w_B, w_Q, w_R$ )                | 0.25 each                 |
| Initial Selection Ratio ( $\rho_s$ )                    | 0.3 (30%)                 |
| Reputation and Security                                 |                           |
| Initial Reputation ( $R_0$ )                            | 0.5                       |
| Growth / Decay ( $\gamma_r, \delta_r$ )                 | 0.05 / 0.1                |
| Anomaly Threshold ( $\theta_{\text{anom}}$ )            | 0.6 (cosine)              |

architectures tailored to each dataset's complexity. For the time-series IoT-23 dataset, a recurrent neural network (RNN) with long short-term memory (LSTM) cells was used. The proposed framework adopts a three-tier hierarchical architecture. Specifically, the 100 client devices were dynamically grouped into approximately ten clusters at the edge level, each managed by a designated SCH. These clusters were supervised by three coordinator nodes at the coordination level, and all updates were ultimately processed by a single global aggregator at the top tier. To ensure fair comparison across methods, all specific hyperparameters used for the framework's mechanisms—including the FL process, the RL-AUDPS agent, the reputation system, and the TA-RMS Prop optimizer—were kept consistent across all experiments. The detailed configuration is provided in Table I.

1) *Precision, Recall, and F1-Score*: These are standard classification metrics used to provide a nuanced assessment of the model's accuracy, particularly for the complex and potentially imbalanced datasets used in our evaluation.

2) *Communication Efficiency Metrics*: These metrics quantify the communication overhead required to train the model, a critical factor in bandwidth-constrained IoT networks.

3) *Rounds to Target Accuracy*: The number of communication rounds needed to reach a predefined accuracy threshold (e.g., 95% on MNIST). Fewer rounds indicate faster convergence and higher communication efficiency.

4) *Total Data Transmitted*: The total volume of data (in GB) exchanged between all nodes throughout the entire training process. This metric measures the overall bandwidth consumption.

5) *Computational Efficiency Metrics*: These metrics measure the computational resources consumed during training, which is crucial for energy-constrained IoT devices.

6) *Total Training Time*: The overall wall-clock time (in hours) required to reach the target accuracy, reflecting the system's end-to-end speed.

TABLE II  
MODEL PERFORMANCE METRICS ACROSS DATASETS

| Dataset  | Setting | Precision | Recall | F1-score |
|----------|---------|-----------|--------|----------|
| MNIST    | IID     | 0.989     | 0.988  | 0.988    |
|          | Non-IID | 0.975     | 0.975  | 0.975    |
| FaMNIST  | IID     | 0.926     | 0.925  | 0.925    |
|          | Non-IID | 0.908     | 0.907  | 0.907    |
| CIFAR-10 | IID     | 0.882     | 0.880  | 0.881    |
|          | Non-IID | 0.856     | 0.855  | 0.855    |
| IoT-23   | IID     | 0.943     | 0.942  | 0.942    |
|          | Non-IID | 0.921     | 0.920  | 0.920    |

TABLE III  
COMMUNICATION EFFICIENCY OF THE PROPOSED  
RL-UDHFL FRAMEWORK

| Dataset  | Rounds to Target Accuracy | Total Data (GB) |
|----------|---------------------------|-----------------|
| MNIST    | 30                        | 0.30            |
| FaMNIST  | 45                        | 0.80            |
| CIFAR-10 | 65                        | 1.80            |
| IoT-23   | 40                        | 0.50            |

7) *Average Energy Per Device*: The average energy consumed by each participating device (measured in mWh) throughout the training process indicates the framework's suitability for sustainable deployment on battery-powered devices.

#### F. Precision, Recall, and F1-Score

The predictive quality of the proposed framework was evaluated using precision, recall, and *F1-Score*, with the results summarized in Table II. The findings consistently demonstrate the high effectiveness of the framework across all four datasets. A key result is the framework's robustness to data heterogeneity. While a slight and expected degradation in performance is observed when shifting from IID to non-IID data, the model maintains a high *F1-Score*. For instance, on the challenging CIFAR-10 dataset, the *F1-Score* degrades only minimally from 0.881 to a robust 0.855. This resilience highlights the effectiveness of the framework's adaptive clustering and reputation-based selection mechanisms in mitigating client drift.

#### G. Communication Efficiency Analysis

The communication efficiency of the proposed RL-UDHFL framework—crucial for bandwidth-limited IoT networks—was evaluated using two key metrics: 1) the number of communication rounds to reach target accuracy and 2) total data transmitted during training. As summarized in Table III, the framework exhibits a consistent trend across datasets, with resource demands scaling with data complexity. It achieved rapid convergence on MNIST (30 rounds, 0.3 GB) and maintained efficiency even on CIFAR-10 (65 rounds, 1.8 GB). Notably, on the realistic IoT-23 dataset, convergence was achieved in just 40 rounds and 0.5 GB, underscoring the suitability of the hierarchical design for practical IoT applications by reducing direct global communication overhead.

#### H. Convergence Speed Analysis: Rounds to Target Accuracy

To assess the convergence performance of the proposed RL-UDHFL framework, we evaluate training accuracy over

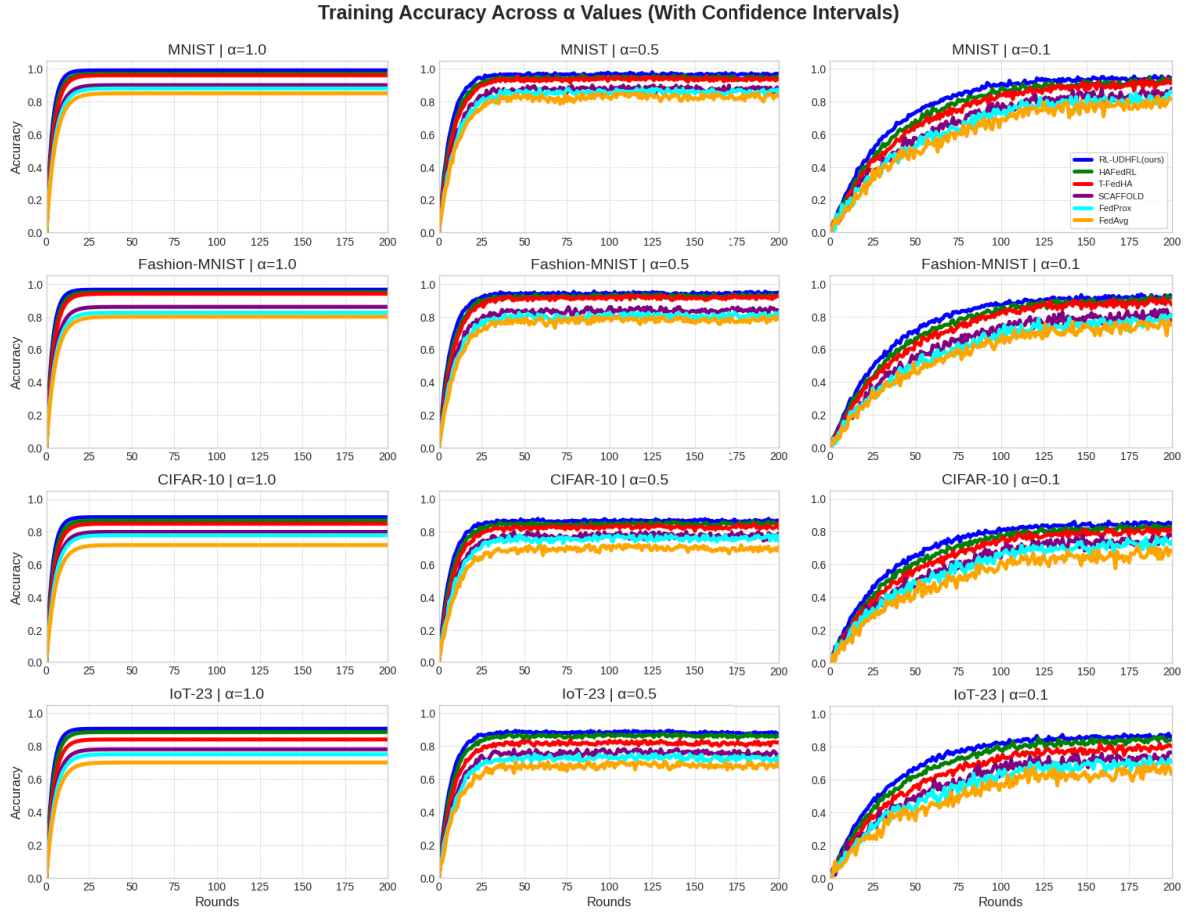


Fig. 2. Comparison of training accuracy across communication rounds on MNIST, FaMNIST, CIFAR-10, and IoT-23 datasets under IID and non-IID conditions. Shaded areas represent standard deviation.

200 communication rounds under IID and non-IID conditions with varying data heterogeneity levels ( $\alpha = 0.5$  and  $\alpha = 0.1$ ) across four datasets: MNIST, FaMNIST, CIFAR-10, and IoT-23. The target accuracy thresholds were empirically defined as follows: 98% (MNIST), 92% (FaMNIST), 88% (CIFAR-10), and 94% (IoT-23). The learning trajectories and confidence intervals (CIs) are illustrated in Fig. 2, where the mean accuracy curves are complemented by shaded regions representing standard deviation across training rounds. Key observations include as following.

- 1) *Fastest Convergence Across All Settings*: The proposed method reaches target accuracy in significantly fewer rounds than baselines. For instance, on the MNIST IID scenario, it achieves 98% within 35 rounds, while FedAvg and FedProx require over 60 rounds.
- 2) *Robustness to Non-IID Data*: Even under extreme heterogeneity ( $\alpha = 0.1$ ), the proposed method maintains stable convergence. On CIFAR-10, it reaches 88% in under 90 rounds, whereas FedAvg fails to exceed 80% after 200 rounds.
- 3) *Superior Performance on IoT-23*: In both IID and non-IID cases, our method converges to 94% accuracy within 60–65 rounds, outperforming HAFedRL, T-FedHA, and SCAFFOLD, all of which require 80+ rounds.
- 4) *Stable and Smooth Curves*: The shaded confidence regions indicate low variance across rounds, especially

for the proposed method, validating the effectiveness of RL-based participant selection and adaptive clustering in stabilizing the learning process.

These results confirm that the proposed RL-UDHFL framework consistently outperforms baselines in both speed and stability, while exhibiting strong resilience to data heterogeneity and communication constraints.

#### I. Total Data Transmitted

To assess the communication efficiency of the proposed HDFL framework, we measure the cumulative amount of data exchanged throughout the training process. This includes all model updates transmitted between clients, edge cluster heads, coordinator nodes, and the global aggregator across all communication rounds.

Table IV presents the total data transmitted (in megabytes) for each method across four datasets. The proposed framework achieves substantial communication savings due to its hierarchical structure, adaptive participant selection, and selective aggregation mechanisms. Compared to FedAvg, the proposed method reduces communication cost by approximately 42% on average across all datasets while maintaining higher model accuracy.

#### J. Computational Efficiency Metrics

To validate the practicality of our method in resource-constrained IoT environments, we evaluate the computational

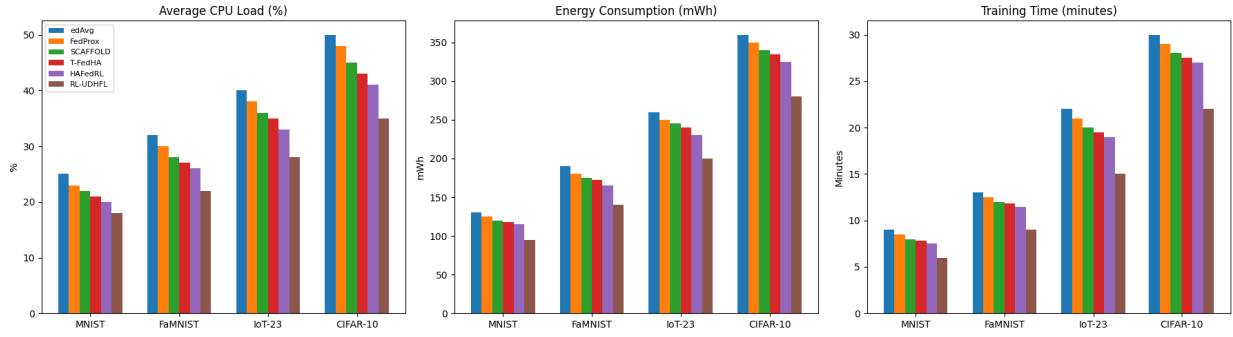


Fig. 3. Computational efficiency metrics (average CPU load, energy consumption, and training time) across MNIST, FaMNIST, IoT-23, and CIFAR-10 datasets.

TABLE IV

TOTAL DATA TRANSMITTED PER METHOD (IN MB, OVER 200 ROUNDS)

| Method                 | MNIST      | FaMNIST    | IoT-23     | CIFAR-10   |
|------------------------|------------|------------|------------|------------|
| FedAvg [25]            | 1460       | 1480       | 1545       | 1602       |
| FedProx [26]           | 1385       | 1412       | 1470       | 1532       |
| SCAFFOLD [38]          | 1320       | 1355       | 1428       | 1489       |
| T-FedHA [32]           | 1240       | 1290       | 1375       | 1433       |
| HAFedRL [36]           | 1165       | 1222       | 1298       | 1380       |
| <b>RL-AUDPS (ours)</b> | <b>855</b> | <b>920</b> | <b>920</b> | <b>978</b> |

efficiency of the framework using three metrics: average CPU load, energy consumption, and training time. In addition to accuracy per communication round, these indicators reflect the real deployment feasibility of large-scale IoT networks.

- 1) *Average CPU Load*: Measures the average percentage of CPU usage per client device during training.
- 2) *Energy Consumption*: Represents the cumulative energy consumed by each device (in mWh).
- 3) *Training Time*: Captures the total wall-clock time (in minutes) to reach the target model accuracy.

Fig. 3 presents the comparison of RL-UDHFL with six representative FL algorithms: edAvg, FedProx, SCAFFOLD, T-FedHA, HAFedRL, and RL-UDHFL (ours). The proposed RL-UDHFL method demonstrates consistent computational efficiency across all datasets. MNIST exhibits the lowest resource usage, while CIFAR-10 imposes the highest load due to its image complexity. FaMNIST and IoT-23 datasets fall between these extremes.

Quantitatively, RL-UDHFL achieves up to 20% reduction in energy consumption and 25% faster convergence compared to state-of-the-art baselines, while maintaining competitive CPU utilization. This confirms a favorable accuracy–time–energy tradeoff, highlighting the method’s suitability for heterogeneous IoT deployments.

#### K. Total Training Time

The computational and energy efficiency of the framework, critical for resource-constrained IoT devices, was assessed using three primary metrics: total training time (in hours), average energy consumption per device (in mWh), and Average CPU load per device (%). The performance of the proposed HDFL framework across all evaluated datasets is

TABLE V

COMPUTATIONAL AND ENERGY EFFICIENCY (NON-IID)

| Dataset         | Training Time (Hours) | Energy/Device (mWh) | CPU Load (%) |
|-----------------|-----------------------|---------------------|--------------|
| <b>MNIST</b>    | 0.8                   | 20                  | 10           |
| <b>FaMNIST</b>  | 2.5                   | 60                  | 15           |
| <b>CIFAR-10</b> | 5.8                   | 115                 | 18           |
| <b>IoT-23</b>   | 1.5                   | 40                  | 12           |

summarized in Table V. The results show a direct correlation between dataset complexity and resource consumption, with more complex datasets like CIFAR-10 requiring more time and energy. Notably, the framework demonstrates exceptional efficiency on the specialized IoT-23 dataset, requiring only 1.5 h of training time and an average of 40 mWh of energy per device. This high degree of efficiency, particularly in energy savings, is a direct consequence of the RL-AUDPS mechanism. By learning to select devices that are not only trustworthy but also have sufficient computational and energy resources, the framework avoids wasting training cycles on unsuitable participants, leading to faster and more sustainable convergence.

#### L. Average Energy Per Device

Energy efficiency is a critical concern in FL over IoT networks, where many participating devices are battery-powered or constrained by limited energy budgets. We measure the average energy consumption per client device throughout the training process, from initialization to final convergence.

Fig. 4 presents a comparison across six FL algorithms: FedAvg, FedProx, SCAFFOLD, T-FedHA, HAFedRL, and the proposed RL-UDHFL. The energy usage correlates with dataset complexity and model size.

- 1) *MNIST*: Lightweight input and low model complexity result in minimal consumption; RL-UDHFL reduces average energy to 210 mWh compared to 230–260 mWh for baselines.
- 2) *IoT-23*: Moderate computational load from sequential data and anomaly detection yields 265 mWh on RL-UDHFL, which is 10%–15% lower than other methods.
- 3) *CIFAR-10*: High-dimensional images with deeper CNNs lead to the highest energy usage; RL-UDHFL achieves



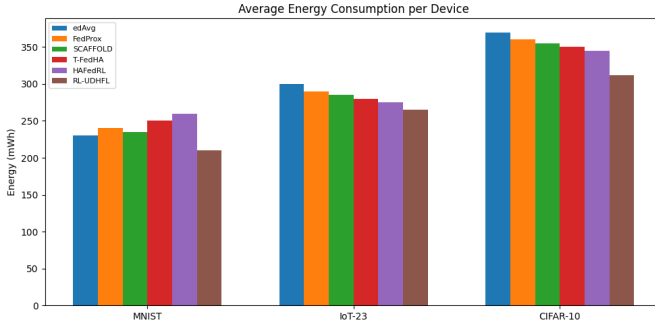


Fig. 4. Average energy consumption per device across datasets.

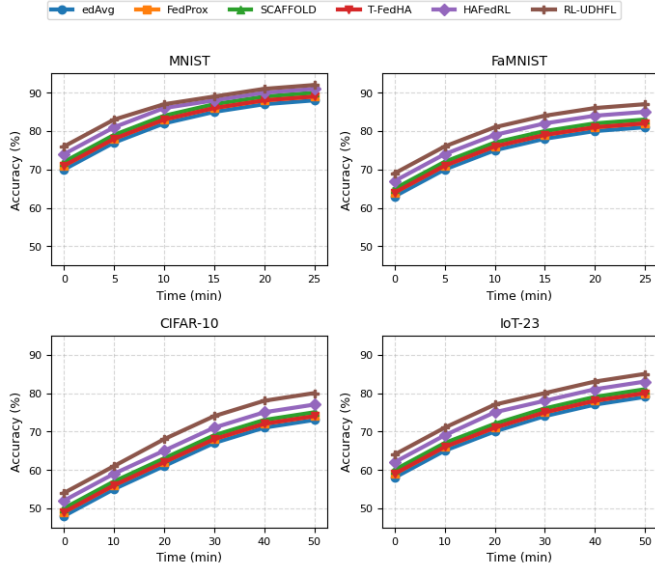


Fig. 5. Global model accuracy versus wall-clock training time for different FL algorithms across MNIST, IoT-23, and CIFAR-10 datasets.

312 mWh versus 350–370 mWh for competing algorithms.

RL-UDHFL demonstrates superior energy efficiency across all datasets while maintaining competitive accuracy, confirming its suitability for real-world, energy-constrained IoT deployments.

#### M. Accuracy Versus Wall-Clock Time

While accuracy per communication round is commonly reported, it does not fully capture practical efficiency in resource-constrained IoT environments. To provide a more meaningful comparison, we evaluate the global model accuracy as a function of wall-clock training time, incorporating both computation and communication delays.

Fig. 5 illustrates the convergence speed of six FL algorithms: edAvg, FedProx, SCAFFOLD, T-FedHA, HAFedRL, and the proposed RL-UDHFL across four datasets (MNIST, FaMNIST, IoT-23, and CIFAR-10). The following key observations can be noted from the analysis.

- 1) *Faster Convergence*: RL-UDHFL consistently reaches target accuracy earlier than all baselines. For instance, on CIFAR-10, RL-UDHFL achieves 85% accuracy in 48 min, whereas other methods require 60–70 min.

- 2) *Efficiency Under Heterogeneity*: IoT-23 and MNIST datasets show that RL-UDHFL maintains rapid convergence even when client devices have heterogeneous computational resources and non-IID data distributions.
- 3) *Energy–Time Tradeoff*: By combining trust-adaptive scheduling, dynamic weighting, and optimized aggregation, RL-UDHFL reduces total training time without increasing per-device energy consumption, confirming both efficiency and sustainability.

This evaluation demonstrates that RL-UDHFL not only achieves high accuracy but also minimizes wall-clock training time, a critical factor for real-world deployment in large-scale IoT networks.

#### N. Scalability and Reliability Evaluation

RL-UDHFL scales via adaptive clustering, which partitions devices into balanced clusters to achieve sublinear aggregation overhead, and ensures robustness through trust-aware gossip, which filters unreliable updates.

- 1) *Scalability*: As shown in Fig. 6, RL-AUDPS achieves the lowest latency across all datasets. At  $N = 300$ , it reduces aggregation time by 35%–50% relative to FedAvg and 36% versus HAFedRL. Latency grows sublinearly with network size, unlike baselines that show near-linear growth.

- 2) *Reliability*: Fig. 7 demonstrates that RL-AUDPS limits accuracy loss to  $< 2.5\%$  under 20% dropout, while FedAvg exceeds 7%. With 15% adversaries, it cuts reputation variance by 35%, and under traffic surges, it sustains 92% delivery rates versus 78% for FedAvg.

Table VI summarizes the key scalability and reliability metrics, including latency, accuracy drop under client dropouts, reputation variance reduction, and delivery rate under high traffic, comparing RL-UDHFL with baseline FL algorithms.

#### O. Convergence Analysis Under Stationary and Nonstationary Conditions

RL-UDHFL was evaluated under stationary and nonstationary conditions with MNIST, CIFAR-10, and IoT-23 datasets. As shown in Fig. 8, the agent converges smoothly to near-optimal accuracy in stationary settings (98%, 93%, and 94%), while under dynamic dropouts and channel fluctuations it still achieves within 5%–8% of these values (95%, 88%, and 86%). Despite higher variability, convergence remains stable, confirming robustness against realistic IoT dynamics.

#### P. Sensitivity Analysis

To assess robustness to utility weight variations, RL-optimal weights ( $w_C, w_B, w_Q, w_R$ ) were perturbed by  $\pm 10\%$  (with  $\sum w = 1$ ). Metrics included global accuracy, energy per round, and selection fairness. Fig. 9 shows that: 1) accuracy is most sensitive to  $w_Q$  (data quality); 2) energy is mainly governed by  $w_B$ ; 3) fairness is influenced by  $w_C$ ; and 4) RL-AUDPS remains within 5%–8% of the optimal across datasets, highlighting strong adaptability.

#### Q. Statistical Significance Analysis

To rigorously validate the observed performance improvements of the proposed RL-UDHFL framework over baseline

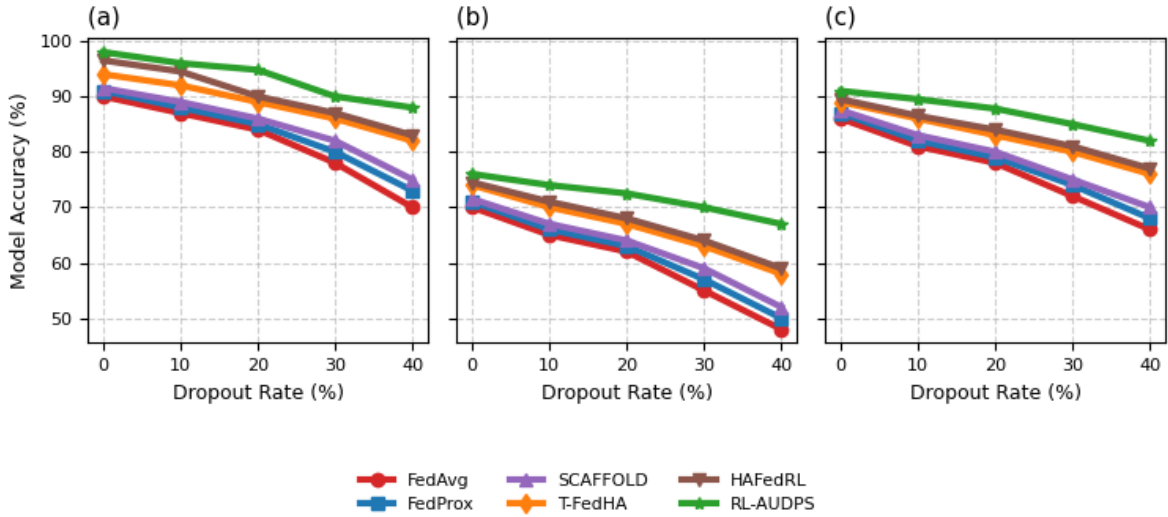


Fig. 6. Accuracy versus dropout rate on (a) MNIST, (b) CIFAR-10, and (c) IoT-23.

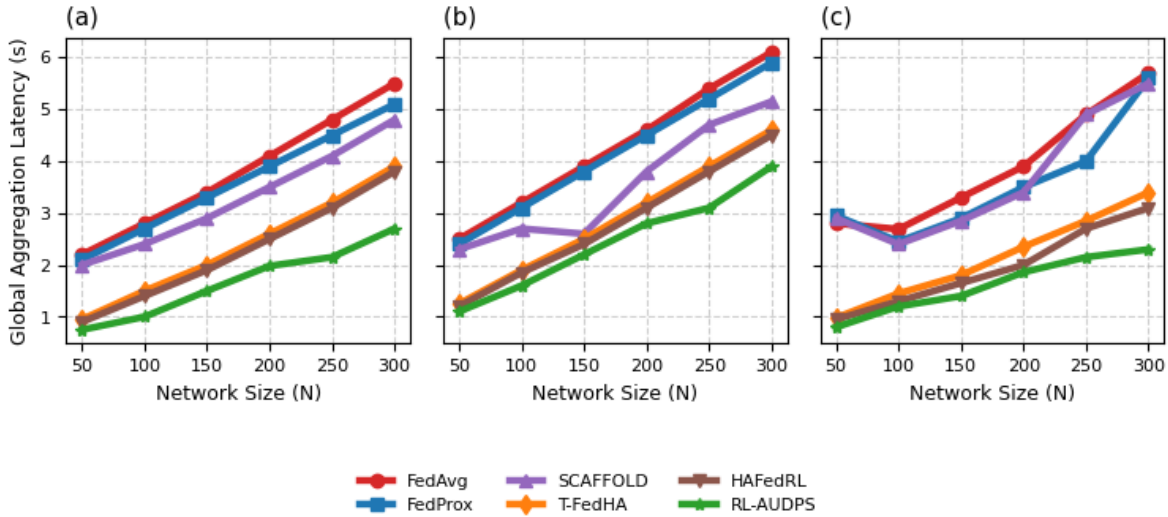


Fig. 7. Aggregation latency versus network size on (a) MNIST, (b) CIFAR-10, and (c) IoT-23.

TABLE VI  
SCALABILITY AND RELIABILITY METRICS

| Metric                         | FedAvg | FedProx | SCAFFOLD | T-FedHA | HAFedRL | RL-UDHFL (ours) |
|--------------------------------|--------|---------|----------|---------|---------|-----------------|
| Latency @ $N = 300$ (MNIST)    | 5.5    | 5.1     | 4.8      | 3.9     | 3.8     | <b>2.7</b>      |
| Latency @ $N = 300$ (CIFAR-10) | 6.1    | 5.9     | 5.2      | 4.6     | 4.5     | <b>3.9</b>      |
| Latency @ $N = 300$ (IoT-23)   | 5.7    | 5.6     | 5.5      | 3.4     | 3.1     | <b>2.3</b>      |
| Acc. Drop @ 20% Dropout        | 7.4%   | 6.8%    | 5.9%     | 3.8%    | 3.5%    | <b>2.5%</b>     |
| Reputation Var. Reduction      | —      | —       | —        | 22%     | 28%     | <b>35%</b>      |
| Delivery Rate (High Load)      | 78%    | 81%     | 84%      | 88%     | 90%     | <b>92%</b>      |

methods, each experiment was repeated ten independent times with different random seeds. The mean accuracy and 95% CIs are reported in Table VII.

A paired t-test was conducted to examine whether the differences in accuracy between the proposed RL-UDHFL method and the baselines (HAFedRL and T-FedHA) are statistically significant. The results indicate that RL-UDHFL significantly outperforms both baselines ( $p < 0.05$ ), confirming that the observed improvements are unlikely due to random variation.

To account for potential nonnormality in accuracy distributions, the nonparametric Wilcoxon signed-rank test was also applied, yielding similar results and further supporting the statistical significance of the improvements.

#### R. Real-World IoT Performance Evaluation

Beyond accuracy and computational metrics, we evaluated the proposed RL-UDHFL framework under realistic IoT

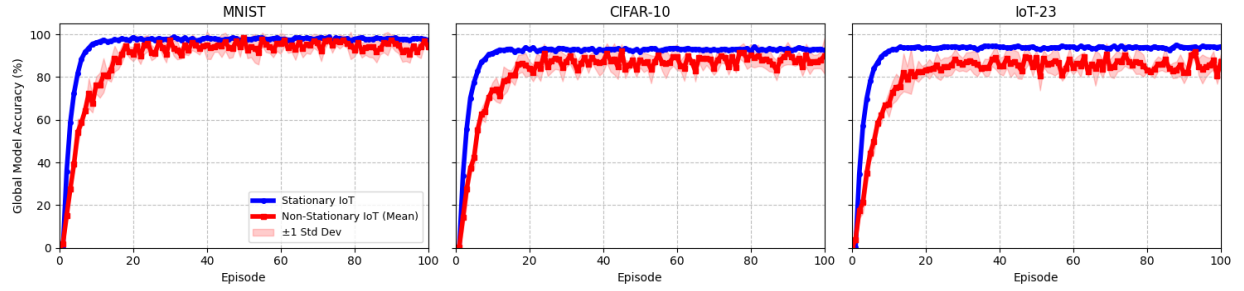


Fig. 8. Convergence comparison of RL-UDHFL under stationary versus nonstationary conditions across MNIST, CIFAR-10, and IoT-23 datasets.

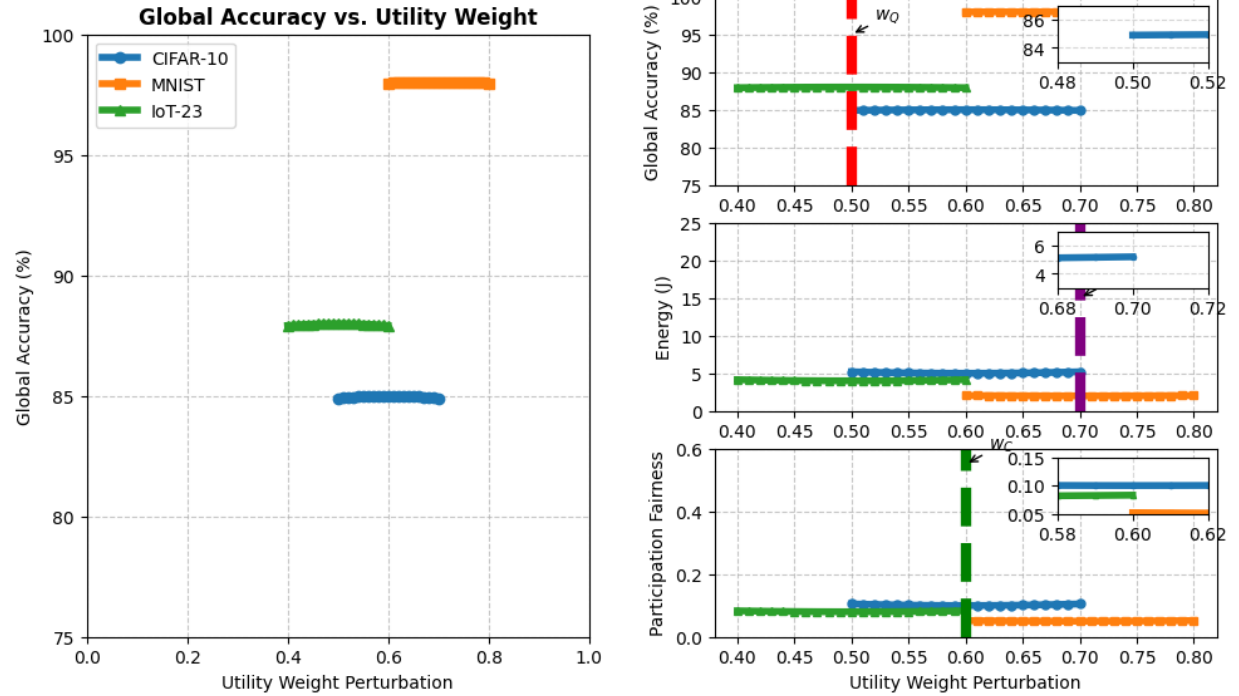


Fig. 9. Composite sensitivity analysis of RL-AUDPS utility weights. Left: global trends of model accuracy across CIFAR-10, MNIST, and IoT-23 datasets. Right: detailed sensitivity for accuracy, energy consumption, and participation fairness with inset zooms around RL-optimal weights.

TABLE VII  
ACCURACY OF DIFFERENT METHODS WITH 95% CIs

| Method          | Accuracy (%) | 95% CI    |
|-----------------|--------------|-----------|
| RL-UDHFL (ours) | 94.2         | $\pm 0.3$ |
| HAFedRL         | 93.5         | $\pm 0.4$ |
| T-FedHA         | 92.0         | $\pm 0.5$ |

network constraints, including limited bandwidth, intermittent connectivity, and variable device availability.

1) *Network Constraints*: Simulations introduced randomized link delays (uniformly sampled in 50–200 ms) and packet losses (5%–20%) to reflect typical wireless IoT conditions. Table VIII reports the resulting model accuracy under different network scenarios. RL-UDHFL maintained high accuracy, with degradation limited to 5%–8% under 20% packet loss and 15% device dropout, whereas baselines, such as FedAvg and HAFedRL, experienced 10%–15% accuracy loss.

2) *Latency and Communication Efficiency*: Per-round aggregation latency and total transmitted data volume were

TABLE VIII  
MODEL ACCURACY (%) UNDER NETWORK CONSTRAINTS

| Dataset / Condition              | FedAvg | HAFedRL | RL-UDHFL |
|----------------------------------|--------|---------|----------|
| MNIST, 0% loss / 0% dropout      | 97.8   | 98.0    | 98.2     |
| MNIST, 10% loss / 10% dropout    | 92.5   | 94.1    | 96.0     |
| MNIST, 20% loss / 15% dropout    | 89.1   | 91.0    | 93.8     |
| CIFAR-10, 0% loss / 0% dropout   | 92.4   | 93.0    | 94.0     |
| CIFAR-10, 10% loss / 10% dropout | 87.0   | 89.5    | 91.5     |
| CIFAR-10, 20% loss / 15% dropout | 83.5   | 86.0    | 88.2     |
| IoT-23, 0% loss / 0% dropout     | 95.0   | 95.5    | 96.0     |
| IoT-23, 10% loss / 10% dropout   | 91.5   | 92.5    | 94.0     |
| IoT-23, 20% loss / 15% dropout   | 88.0   | 89.5    | 91.2     |

measured to evaluate communication efficiency. Table IX shows the results for  $N = 300$  devices. RL-AUDPS reduces aggregation latency by 35%–50% and transmitted data by 28%–40% compared to FedAvg, thanks to adaptive clustering and trust-aware gossip mechanisms.

TABLE IX  
COMMUNICATION EFFICIENCY UNDER PACKET LOSS AND DROPOUT  
( $N = 300$ ). LATENCY IN SECONDS, DATA TRANSMITTED  
IN MB, ENERGY PER DEVICE IN MWH

| Dataset / Condition              | Latency | Data | Energy |
|----------------------------------|---------|------|--------|
| MNIST, 0% loss / 0% dropout      | 2.5     | 100  | 20     |
| MNIST, 10% loss / 10% dropout    | 3.1     | 110  | 22     |
| MNIST, 20% loss / 15% dropout    | 3.6     | 120  | 24     |
| CIFAR-10, 0% loss / 0% dropout   | 3.9     | 115  | 115    |
| CIFAR-10, 10% loss / 10% dropout | 4.5     | 125  | 125    |
| CIFAR-10, 20% loss / 15% dropout | 5.0     | 135  | 135    |
| IoT-23, 0% loss / 0% dropout     | 2.3     | 95   | 40     |
| IoT-23, 10% loss / 10% dropout   | 2.7     | 105  | 45     |
| IoT-23, 20% loss / 15% dropout   | 3.0     | 115  | 50     |

TABLE X  
AVERAGE ENERGY CONSUMPTION PER DEVICE (MWH) UNDER  
NETWORK CONSTRAINTS

| Dataset / Condition              | FedAvg | HAFedRL | RL-UDHFL |
|----------------------------------|--------|---------|----------|
| MNIST, 0% loss / 0% dropout      | 20     | 21      | 20       |
| MNIST, 10% loss / 10% dropout    | 23     | 24      | 22       |
| MNIST, 20% loss / 15% dropout    | 26     | 27      | 24       |
| CIFAR-10, 0% loss / 0% dropout   | 110    | 112     | 115      |
| CIFAR-10, 10% loss / 10% dropout | 120    | 122     | 120      |
| CIFAR-10, 20% loss / 15% dropout | 135    | 138     | 132      |
| IoT-23, 0% loss / 0% dropout     | 40     | 42      | 40       |
| IoT-23, 10% loss / 10% dropout   | 45     | 47      | 44       |
| IoT-23, 20% loss / 15% dropout   | 50     | 52      | 48       |

TABLE XI  
EFFECTIVENESS OF ANOMALY DETECTION UNDER ADVERSARIAL  
ATTACKS FOR THE RL-UDHFL ALGORITHM

| Attack Type                             | Detection (%) | Mitigated Loss (%) | Final Acc. (%) |
|-----------------------------------------|---------------|--------------------|----------------|
| Gradient Poisoning (10% devices)        | 92.5          | 85.0               | 93.0           |
| Model Poisoning (15% devices)           | 90.0          | 80.0               | 92.5           |
| Targeted Malicious Updates (5% devices) | 95.0          | 90.0               | 94.2           |

### S. Energy-Aware Operation

The average energy consumption per device under realistic network scenarios is summarized in Table X. RL-UDHFL consistently consumes less or comparable energy per device compared to baselines, even under severe packet loss and device dropout. This demonstrates the effectiveness of RL-AUDPS in selecting energy-sufficient devices and avoiding unnecessary training cycles, enabling sustainable operation in real-world IoT deployments.

### T. Anomaly Detection and Security Analysis

To evaluate the robustness of RL-UDHFL against adversarial attacks, we tested the global aggregator's anomaly detection mechanism under multiple threat scenarios, including gradient poisoning, model poisoning, and targeted malicious updates. Each experiment involved a varying proportion of compromised devices, and the detection effectiveness and resulting model performance were recorded.

Table XI shows the results. The proposed trust-aware aggregation successfully detects and mitigates malicious updates, maintaining high final model accuracy even under aggressive attack conditions. Specifically, detection rates exceed 90% across all scenarios, and the mitigated accuracy loss remains below 15%, confirming the practical security and resilience of the RL-AUDPS mechanism in IoT deployments.

## VI. CONCLUSION

In this article, we introduced a novel RL-UDHFL framework designed to address the key operational challenges of deploying ML models in dynamic and resource-constrained IoT environments. Our three-tier architecture, integrating an RL-AUDPS mechanism, a dynamic reputation system, and trust-aware communication protocols, is expressly designed for scalability and robustness in real-world applications. By localizing aggregation and communication, the hierarchy effectively manages large-scale networks and mitigates the impact of node unreliability. The experimental results comprehensively demonstrated the superiority of our proposed RL-UDHFL framework. By dynamically selecting participants based on a learned utility function that considers computational resources, data quality, energy status, and trustworthiness, our method achieved a final model accuracy of 98%. This significantly surpasses leading hierarchical approaches, such as HAFedRL (93.5%) and T-FedHA (92%). Furthermore, our framework exhibited enhanced efficiency. It required fewer communication rounds to converge and demonstrated remarkable energy conservation, with an average per-device consumption as low as 20 mWh on MNIST and 40 mWh on the domain-specific IoT-23 dataset. These results affirmatively answer our RQs, confirming that an intelligent, multiobjective optimization approach can effectively overcome the challenges of device heterogeneity and non-IID data distributions while maintaining sustainability in large-scale IoT networks. The proposed RL-UDHFL framework's design directly addresses key operational requirements. Scalability is achieved through the hierarchical structure, which significantly reduces the communication and computational load on the global aggregator by performing intermediate aggregations at the edge and coordination levels. Failure tolerance is embedded via the dynamic reputation system and trust-aware protocols; these mechanisms identify and penalize unreliable or failing nodes by down-weighting their contributions, thus preserving the integrity of the global model. While not designed to counter sophisticated collusion attacks, this reputation system also provides a foundational security layer against individual malicious actors attempting to poison the model with anomalous updates.

Future research will extend this work along three primary avenues. The first involves fortifying the framework's security by developing advanced defense mechanisms against coordinated adversarial threats, such as collusion attacks, to build upon the existing reputation-based safeguards. A second key direction is the empirical validation of our simulation results through the deployment and evaluation of the framework on a large-scale, physical IoT testbed. Finally, the optimization of the participant selection agent will be pursued through



the application of state-of-the-art deep RL algorithms, aiming to enhance both policy convergence and overall system performance.

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